NSF UIC RET Curricular Alignment Summary for AP Statistics
(Aligned to 2026-27 Course and Exam Description)

The 8-lesson series developed during this 6-week Research Experience for Teachers addresses approximately 62% of the 2026-27 AP Statistics Learning Objectives (LOs). The curriculum uses water infrastructure scenarios to produce data that takes students from descriptive analysis through probability and simulation, inferential statistics (Chi Square Tests) and ends with regression modeling. By utilizing the SWIM Lab's computational modeling and simulation (CMS) tools, students engage with 36 of the approximately 58 unique Learning Objectives defined by the College Board.

Breakdown of CED Coverage by Unit

CED Unit	Coverage	Key Learning Objectives (LOs) Addressed
Unit 1: Exploring One-Variable Data and Collecting Data	**~90%**	1.1.A/B (Investigative Questions), 1.2.B (Variables), 1.6.A (Descriptions), 1.7.A/B/D/F (Summary Stats/Outliers), 1.9.A/B/C (Comparisons), 1.10.A/D/E (Study Design/Confounding).
Unit 2: Probability, Random Variables, and Distributions	**~85%**	2.1.A (Two-way Tables), 2.3.A (Simulation), 2.4.A (Probability), 2.6.A (Conditional Prob), 2.7.A/B (Independence), 2.9.A/B (Parameters), 2.10.A/B/C (Binomial).
Unit 3: Inference for Categorical Data: Proportions	**~15%**	3.14 (Chi-Square Setup) and 3.15 (Chi-Square Execution). Focus is on the relationship between categorical variables in infrastructure safety.
Unit 4: Inference for Quantitative Data: Means	**~10%**	4.5.A (Statistical vs. Practical Significance). Concepts of variability and mean shifts are introduced in the context of sensor optimization.
Unit 5: Regression Analysis	**~100%**	5.1.A (Scatterplots), 5.2.A (Correlation), 5.3.A (Predictions), 5.4.A (Residuals), 5.5.A (Slope/Intercept), 5.5.B (Coefficient of Determination).

Curricular Gaps and Future Development

During the 6-week research experience I prioritized the development and validation of the SWIM Lab computational modeling lessons, which required substantial time for simulation design, data generation, pedagogical reasoning, and classroom integration. Developing curricular ideas was relatively easy and afforded me the opportunity to become more familiar with the new AP Statistics CED. Revising the Python code was the most time-consuming part of the experience. Because of the time constraints, the formal inference sequences for Units 3 (inference for proportions) and 4 (inference for means) were not developed during the six-week period. The plan is to develop these components during the 2026-27 academic year, building upon the simulation-based foundation and real-world data structures established through this work.

Area Not Covered	Rationale for Omission in RET
Unit 3: Proportion Inference (Most of it)	The focus prioritized categorical analysis via contingency tables and Chi-Square tests to model system-wide safety status. Tests involving z-distributions were reserved for later development
Unit 4: Means Inference	While "Practical Significance" was explored with Chi Square Tests, formal t-distribution theory was reserved for later development.

Research Experiences for Teachers
University of Illinois at Chicago
Smart Water Infrastructure Modeling (SWIM) Lab
Overview of AP Statistics Lessons Developed

Lesson 1: Formulating Questions to Guide Statistical Investigations
Describing, Comparing and Summarizing Quantitative Distributions (Desmos & Excel)
CMS Scenario: Normally operating water infrastructure. Comparing end-nodes and through-nodes pressure, as well as chlorine levels. Methods to identify outliers. Introduce concept of confounding variables.

Lesson 2: Continuation of Lesson 1 concepts, but introduce impact of skewness on mean, median, standard deviation and IQR. Introduce concept of statistical significance at a high-level.
CMS Scenario: Randomly select a critical pipe (based on diameter). Introduces students to random selection. Assess the impact of that pipe closure on pressure at other junctions in the network. Classify a junction that experiences greater than or equal to 5 PSI decrease as vulnerable, otherwise classify it as resilient. Introduces students to categorical nature of some quantitative variables. Compare chlorine level and water age in resilient vs. vulnerable zones.

Lesson 3: Formative assessment on the concepts covered in the 1st 2 lessons.
CMS Scenario: Randomly select a critical pipe (based on diameter). Introduces students to random selection. Students analyze the impact of that pipe closure on the mixture of lake vs river water in the system. They compare graphical displays (boxplot vs histogram) and the advantages/disadvantages for displaying the data.

Lesson 4: Data analysis with a focus on the mean and standard deviation. Establishes what "information" the mean and standard deviation provide to a data analyst. What constitutes a chronic problem vs. an acute problem? How can that information help with sensor placement.
CMS Scenario: Event 1 high-volume fire flow (hours 48 – 54), pump maintenance (hours 96 to 108) residential fire lower-volume fire flow. Excel file produced with mean and standard deviation of pressure, water age and chlorine levels. Students analyze the raw data to make decisions on sensor placement. Class will identify 3 junctions that are candidates for sensor placement. Display time-series analysis graphs to decide if that placement was a "good" choice given the longitudinal data.

Lesson 5: Random Variables, Probability Distributions, Contingency Tables, Conditional Probability, Independent Events. Students build a contingency table from raw data. Students use that table to find both marginal and conditional probabilities and determine if the response variable (water safety) is independent of the explanatory variable (junction zone – resilient vs vulnerable).
CMS Scenario: Pipe closure for 24 hours. Same scenario as earlier, select a critical pipe at random and close it. Randomly select a vulnerable junction (if any) and resilient junction, measure the chlorine level, classify it as safe or unsafe. Then re-open the pipe, run the simulation for 2 more hours, classify the randomly selected junctions accordingly.

Lesson 6: Binomial probabilities, discrete random variables, classifying a variable as binomial, binomial probability calculations.
CMS Scenario: Familiar simulation scenario, close a pipe, run the infrastructure for 24 hours and classify junctions as safe or unsafe (based on chlorine levels). This scenario asks the user to enter how many junctions to randomly select and how many simulated trials to run. It produces output that includes how many of the XX junctions were unsafe. It also provides an empirical probability of being unsafe as well as the mean number of unsafe junctions from the simulated trials.

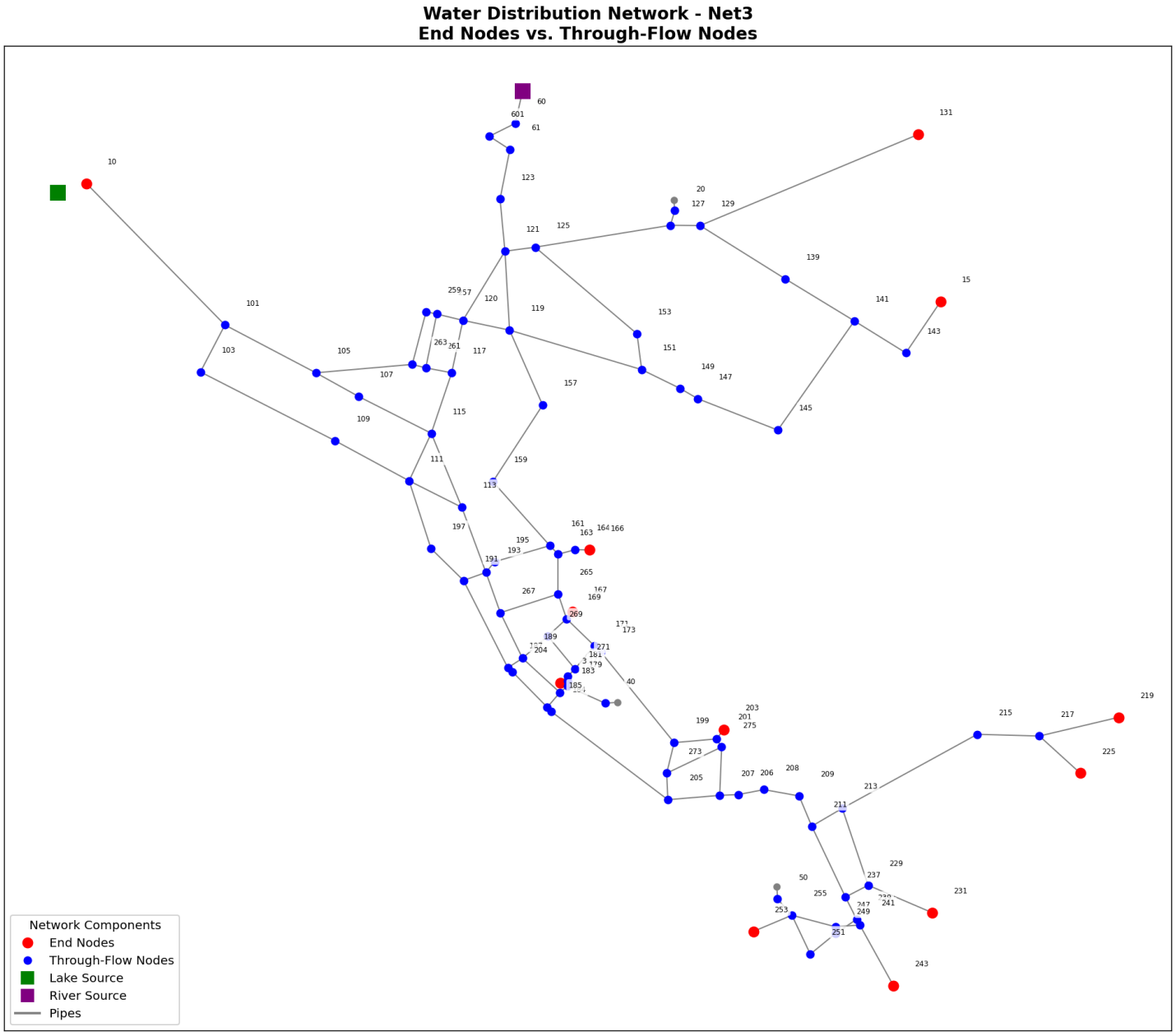
Lesson 7: Contingency tables, conditional probability, independent events, Chi Square Test of Independence**.** Is the type of junction (based on number of connections) related to water safety when a contaminant enters the system?
CMS Scenario: Introduce a contaminant at the river source that goes undetected and spreads for 6 hours. Close the pipe from the river source when the contamination is detected. Go another 12 hours while the reservoir is drained. The reservoir and its river source is now clean, but the contaminated water is still in the pipes. The reservoir pipes are now re-opened so that clean water enters the system. Run the system for 2 more weeks. At each stage, contingency tables are built with 6x1 hour data collection at each junction. Connections are made between earlier optimal sensor placement and the new information we gain from this analysis. Connections are made between Chi-Square Test of Independence and conditional probability calculations. Practical vs Statistical significance is introduced as well.

Lesson 8: Scatterplots to display relationships between two quantitative variables, least-squares regression lines, using residuals to assess the appropriateness of a model, interpreting slope and y-intercepts in context, interpreting correlation and the coefficient of determination, cautions when extrapolating.
CMS Scenario: Run the system as normal, with no stressors. Collect a snapshot of the infrastructure with readings from each of the 92 junctions. Introduce students to scatterplots and regression analysis with explanatory variable water age and response variable chlorine level as well as explanatory variable elevation and response variable pressure. It is expected that fairly strong negative linear relationships will exist. After 24 hours, a pipe is closed. No labeling of resilient or vulnerable junctions are done in this simulation. After 48 hours (24 hours after pipe is closed), another snapshot is take of the system. NOTE: this is not a time-series analysis, it is a snapshot of a system under stress. Students analyze if the model under stress and the model under normal conditions are similar. This is an introduction to extrapolation and reasons to be cautious. The Excel file also contains the 2 snapshots combined (184 junction readings). They should notice two distinct patterns with the elevation vs pressure. They should also notice a negative pressure reading at the highest junction. This is indicative of a system failure (more demand than the system can meet. The model that combines the 2 snapshots should show a very weak linear relationship for water age vs. chlorine level, which should lead to meaningful discussions.

AP Statistics **Describing Data Numerically & Graphically**

Sirois

End Nodes vs. Through-Flow Nodes



Background: The Hydraulics of Network Topology

In a water distribution network, water pressure is a critical metric for system health. Pressure is influenced by many factors, including elevation, pipe diameter, and the "topology" (layout) of the network. In this investigation, we will compare two distinct types of junctions in the **Net3** infrastructure:

1. **End Nodes (Dead-Ends):** Junctions connected to only one pipe. These are the "cul-de-sacs" of the water system.
2. **Through-Flow Nodes:** Junctions connected to two or more pipes. These are the "intersections" where water passes through to reach other parts of the city.

The Investigative Question #1: Does the position of a junction in the Net3 water distribution network (End Node vs. Through-Flow) impact the water pressure experienced at that junction type? Before trying to answer this question, let's look at the 3 important criteria that should be present in any investigative question for a statistical study.

A. Does this question direct the type of data and variables that we will collect?
   Yes. We have 2 categories that we will need to separate the data into, End Nodes and Through Flow Nodes. We will collect water pressure at each of those nodes.
B. Does the question inform what type of analysis will be done?
   Yes. We will compare distributions of each type of node using the Shape, Outlier, Center and Spread (SOCS) approach. This means we will need to calculate numerical summaries for the data and also construct a graphical summary (we'll try both boxplots and histograms using Desmos as a tool).
C. Does the question indicate the population these results can be generalized to?
   Yes. The population is identified as the Net3 water distribution network

Data has been provided for today's lesson in an Excel file. The link is on the Topics Page for today's lesson.

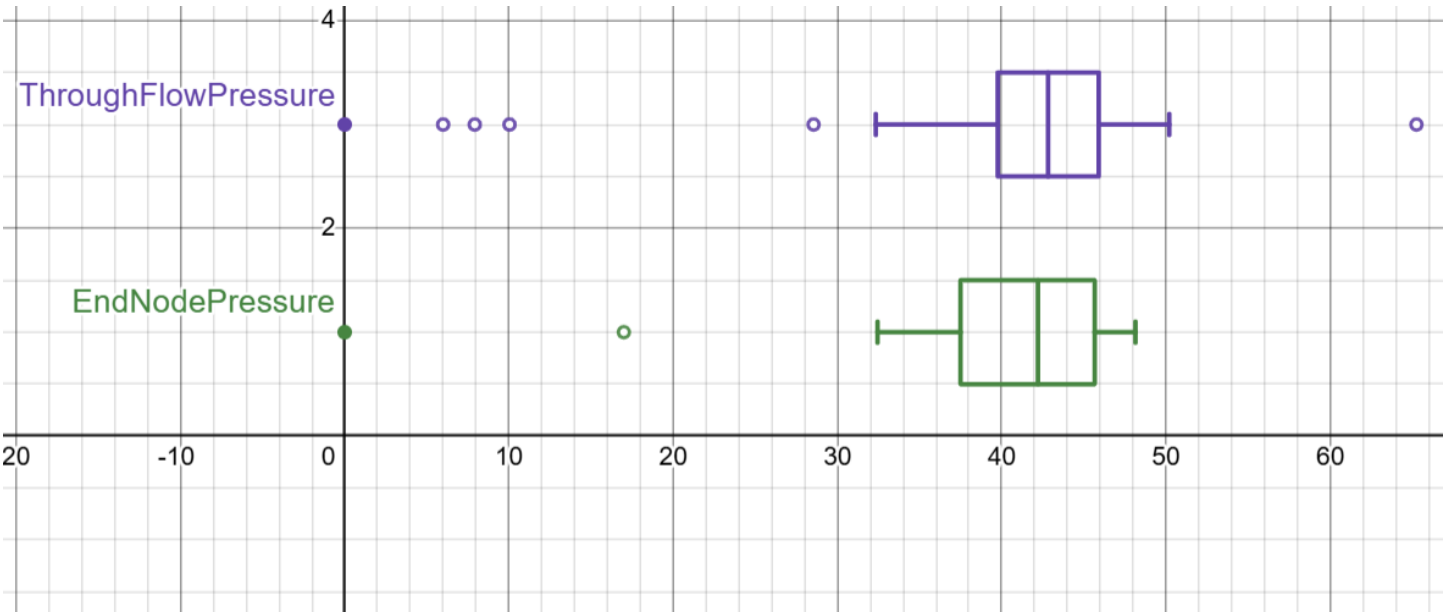
We will use that data, and learn to copy & paste it into Desmos to help with our data analysis.

Analysis Questions:
1. Based on the boxplots, which measures of center and spread should we use? Justify your choice by explaining which measures are resistant to outliers and skewness. Based on the boxplots, which group (if any) appears to have a higher typical pressure? Justify your answer using a specific statistic.
2. Compare the variability (spread) of the two distributions. Which group has more consistent pressure? Which measure of variability are you using? Justify your choice.
3. Are there any outliers present in either distribution? Use the 1.5×IQR rule to locate the lower and upper fence for each distribution of pressure readings.

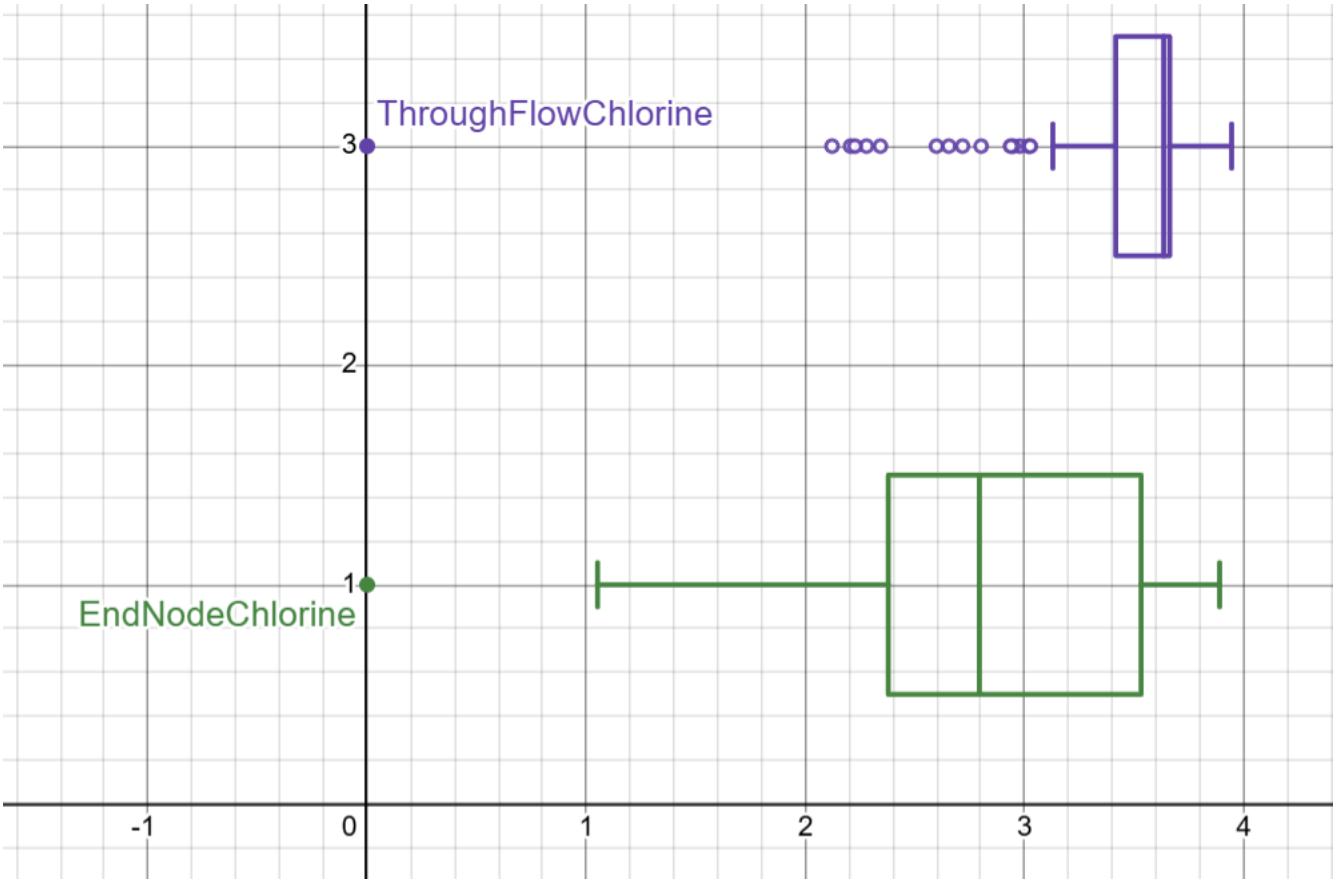
The Investigative Question #2: Does the position of a junction in the Net3 water distribution network (End Node vs. Through-Flow) impact the chlorine level experienced at that junction type? Use the same 3 analysis questions from above to formulate your answer.

Confounding Variable Introduction
A difference in pressure between these two groups might not be caused only by their position in the network. There may be confounding variables. A variable is considered confounding if it affects the response variable AND is associated with the explanatory variable, making it difficult to determine whether the response is due to the explanatory variable or to the confounding variable itself. For example, if end nodes tend to have smaller pipe diameters than through-flow nodes, then pipe diameter could be a confounding variable since it is associated with the explanatory variable (node type) and has a direct impact on the response variable (bigger diameters are associated with higher pressure).

Statistical Question Writing: Based on your analysis today, write one new statistical question that could be investigated using this water infrastructure model. Ensure your question guides data collection and analysis. Consider the scenario in which a critical pipe in the infrastructure fails. Use that scenario to formulate your question. You want to determine if the closure of the critical pipe significantly degrades the safety of the water at a specific junction in the network. Remember: A good statistical question should identify the variable(s) of interest, specify the population, and guide the type of analysis. Submit your question before leaving class today.



Opportunity to introduce outlier test, numerical summaries, etc. With Chlorine data, introduce idea that chlorine levels may be statistically significantly higher in through-flow pipes. Introduce confounding. Introduce practical significance. Compare mean to median with ThroughFlow pipe chlorine levels as well as above.



AP Statistics Curriculum and Exam Description (CED) Alignment

Unit 1: Describing Data Numerically and Graphically

The following Learning Objectives (LO) and Essential Knowledge (EK) components from the 2026-27 AP Statistics Course and Exam Description are addressed in the "End Nodes vs. Through-Flow Nodes" case study.

Section 1: Formulating Investigative Questions

- **LO 1.1.B:** Determine an investigative question within a statistical study.
- **EK 1.1.B.2:** An investigative question should be posed so that the required data can be collected and analyzed.
- **LO 1.10.A:** Determine the components of an investigative question within a statistical study.
- **EK 1.10.A.1:** The first component of an investigative question should guide the data collection process and should be phrased in terms of the variable(s) of interest in the study.
- **EK 1.10.A.2:** The second component of an investigative question should guide the data analysis choice.
- **EK 1.10.A.3:** The third component of an investigative question should indicate the type(s) of conclusion(s) applicable from the study. The investigative question should provide the population to which the conclusions will be applicable.

Section 2: Identifying Variables and Populations

- **LO 1.1.A:** Identify components within a statistical study.
- **EK 1.1.A.4:** A population consists of all items or individuals of interest.
- **LO 1.2.B:** Identify types of variables.
- **EK 1.2.B.1:** A categorical variable, also called a qualitative variable, takes on values that are category names or group labels.
- **EK 1.2.B.2:** A quantitative variable, also called a numerical variable, takes on numerical values for a measured or counted quantity and generally has units of measure.

Section 3: Analyzing and Describing Distributions

- **LO 1.6.A:** Describe distributions of quantitative one-variable graphical representations.
- **EK 1.6.A.1:** Descriptions of the distribution of one quantitative variable include shape, center, and variability (spread) as well as any unusual features such as outliers, gaps, or clusters in context.
- **LO 1.7.F:** Justify the selection of a summary statistic for describing quantitative data.
- **EK 1.7.F.1:** The median and IQR are considered a resistant (or robust) measure of center and measure of variability, respectively, because outliers do not greatly (if at all) affect their values.
- **LO 1.9.A:** Compare multiple quantitative one-variable graphical representations.

Section 4: Outlier Detection

- **LO 1.7.D:** Calculate outliers for quantitative data.
- **EK 1.7.D.1.i:** An outlier is a value located more than $ 1.5 \times IQR $ above the third quartile or more than $ 1.5 \times IQR $ below the first quartile.

Section 5: Observational Studies and Confounding

- **LO 1.10.D:** Identify an observational study.
- **EK 1.10.D.5:** A confounding variable in an observational study provides an alternative explanation for the observed relationship... To be a confounding variable, a variable must be associated with both the explanatory variable and the response variable.

Section 6: Synthesis and Question Writing

- **LO 1.A:** Determine a valid investigative question that requires a statistical investigation.
- **LO 1.10.A:** Determine the components of an investigative question within a statistical study.

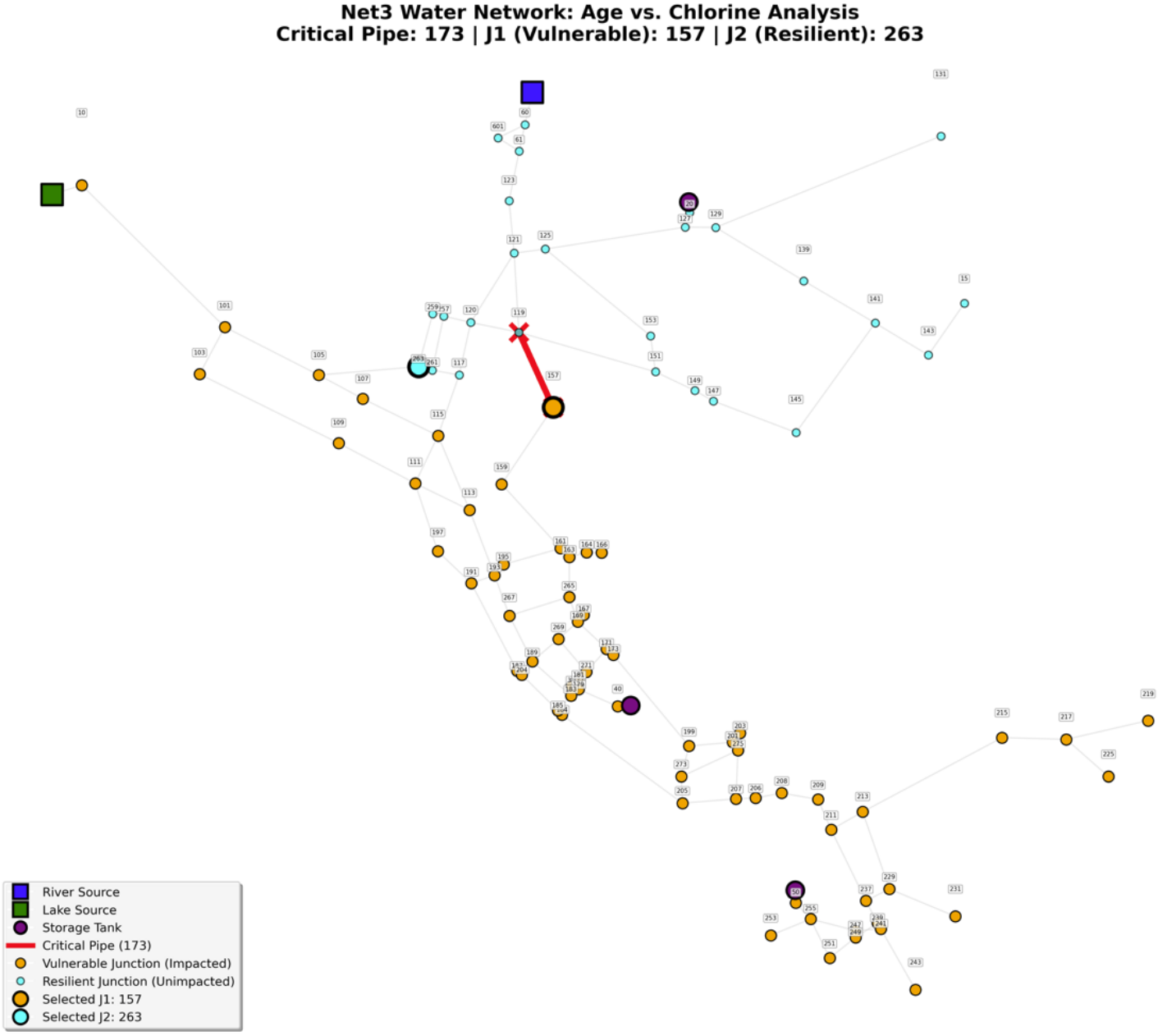
AP Statistics **Impact of Pipe Closure on Water Quality Lesson Set**
Sirois **UIC - SWIM Lab (Smart Water Infrastructure Modelling)**

Overview

This activity uses simulated data from the **WNTR (Water Network Tool for Resilience)** library to explore how infrastructure failures impact public health. You will act as a researcher at the **UIC SWIM Lab**, investigating how a "Critical Pipe" closure affects a city's water safety.

Background: The Physics of the System

A water network is a complex web of interconnected pipes. Some pipes are "critical" because they serve as the primary highway for water moving from the treatment plant (the **Reservoir**) to the rest of the city.



In our simulation, we identify the top 10 most critical pipes in the system, randomly select 1 of the 10 and "close" it. This represents a catastrophic main break or an emergency maintenance shutdown. When this highway is blocked, the water is forced to find alternative paths (detours) through smaller, secondary pipes to reach residents.

Key Water Quality Variables:

1. **Water Age:** The time it takes for a drop of water to travel from the treatment plant to a specific junction. Older water is more likely to grow bacteria.
2. **Chlorine Concentration:** A disinfectant added at treatment plants to kill pathogens. It decays over time; as water age increases, chlorine levels decrease.

Safety Standard: The EPA recommends a chlorine residual between 0.2 mg/L and 4.0 mg/L.

Part 1: Formulating Investigative Questions

A strong investigative question is the foundation of any statistical study. A complete question must include three components:

- **Component 1:** Guide the data collection (identify variables of interest).
- **Component 2:** Guide the data analysis (specify the parameter and direction of the alternative hypothesis or goal of estimation).
- **Component 3:** Indicate the type of conclusion (specify the population and whether cause-and-effect is applicable).

Analyzing The Impact of Infrastructure Failure – Formulating an Investigative Question

Scenario: You need to determine if the closure of the critical pipe significantly degrades the safety of the water at a specific junction in the network.

Step 1: Draft Your Initial Question Based on the scenario, write a statistical question that would help you investigate the impact of the pipe closure on water safety.

Step 2: Data Collection Question Audit If you were going to answer your question above, what specific data would you need to request from the SWIM Lab simulation?

- What variable(s) must be measured?
- What groups or conditions must be compared?
- What specific population are you studying?

Step 3: Self-Assessment Checklist Check your draft against the three required components. If you miss one, revise your question!

- **Component 1:** Did I name the specific variable (e.g., Chlorine) and the groups (e.g., Normal vs. Closed)?
- **Component 2:** Did I specify a parameter and a direction?
- **Component 3:** Did I state the population that the results could be generalized to?

Step 4: Peer Feedback & Revision Trade papers with a partner. Provide feedback using the following prompts:

- I notice you included...
- I wonder if your question would be clearer if...
- I suggest you strengthen Component [1, 2, or 3] by...

Step 5: Final Investigative Question

Write the final form of your investigative question.

Analyzing The Impact of Infrastructure Failure
Describing and Comparing Distributions to Help Answer a Question

Now that you have formulated your questions, you have been provided with data that resulted from a SWIM Lab Simulation.

Using Desmos to help construct graphs that will help analyze the data
You have been provided with a spreadsheet that contains data based on the type of zone (vulnerable vs. resilient). The variables measure the age of the water as well as the chlorine level. Graphs of data help us to describe and compare important features of numerical distributions. Numerical summaries also aid in those descriptions.

Use this time to practice constructing both a boxplot and a histogram for the age of water measured in vulnerable areas both before and after the critical pipe was closed. What features of the distribution do you notice in a histogram that are not present in the boxplot?

Water Age for the following Conditions	Mean ($\bar{x}$)	Std. Dev (s)	Min	Q1	Median	Q3	Max
Vulnerable Normal							
Vulnerable Closed							
Water Age for the following Conditions	Mean ($\bar{x}$)	Std. Dev (s)	Min	Q1	Median	Q3	Max
Resilient Normal							
Resilient Closed							

Describing the Distribution (Vulnerable Zone Closed Age)
Using your boxplot, histogram and numerical summaries to describe the distribution of water age when the pipe is closed. Remember to include shape, outliers, center and spread, all in context)

1. Compare the distribution of the age of the water in the vulnerable zone to the resilient zone.

Variable 2: Chlorine Concentration
Chlorine is a disinfectant added at the treatment plant to kill bacteria and viruses. However, chlorine is a reactive chemical; it naturally **decays** (disappears) over time as it sits in the pipes.

2. Compare the chlorine level in the water in vulnerable and resilient zones.

Confounding Revisited
In statistics, a confounding variable is an outside influence that is related to both the explanatory and response variables. In our study of the water network, we are investigating if a Pipe Closure (Explanatory Variable) causes a change in Chlorine Levels (Response Variable) or the Water Age (another Response Variable).

However, other factors might be provide alternative explanations for why some junctions have lower chlorine than others. The goal of this part of the lesson is to provide a context for confounding, but also making sure you understand that confounding variables are related to both the explanatory and response variables and those relationships make it difficult to determine what impact the explanatory variable actually had on the response.

1. Distance from the Source (Hydraulic Path)

- **Why it could be confounding:** Junctions classified as "Vulnerable" might simply be located further away from the treatment plant (Reservoir) than "Resilient" junctions.
- **The Logic:** Even under normal conditions, water traveling to distant nodes takes longer to arrive, allowing more time for chlorine to decay. If our vulnerable junctions are naturally far away, the low chlorine might be due to their location rather than the pipe closure itself.

2. Pipe Diameter Distribution

- **Why it could be confounding:** The size of the pipes feeding a junction significantly impacts the velocity of the water.
- **The Logic:** Smaller pipes generally have slower flow rates, which increases water age. If the "Vulnerable Zone" is primarily served by a network of small-diameter secondary pipes, while the "Resilient Zone" is served by large trunk mains, the difference in chlorine could be a result of infrastructure design rather than the emergency shutdown.

3. Local Demand Levels (Consumption Rates)

- **Why it could be confounding:** Water age is heavily dependent on how quickly residents are actually using the water.
- **The Logic:** In high-demand areas, water is constantly being pulled through the pipes and replaced with fresh, highly-chlorinated water from the source. In low-demand areas, water sits stagnant for longer periods. If our two comparison groups have different average demand levels, the "demand" becomes a confounding variable that masks the true effect of the pipe closure.

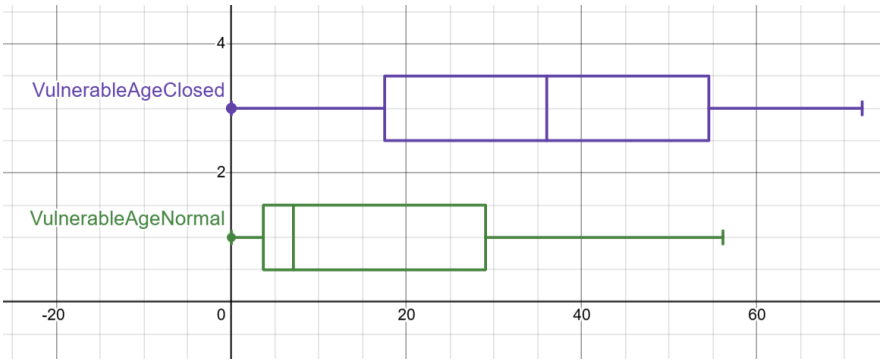
4. Proximity to Storage Tanks

- **Why it could be confounding:** Storage tanks act as buffers in a water network, holding large volumes of water to meet peak demand.
- **The Logic:** Water stored in a tank has its own "age" profile. Junctions located immediately downstream of a tank may receive water that has been sitting in storage for 24–48 hours. If one of our study groups is closer to a tank than the other, the tank's storage cycle—not the pipe closure—might be the primary driver of the observed chlorine levels.

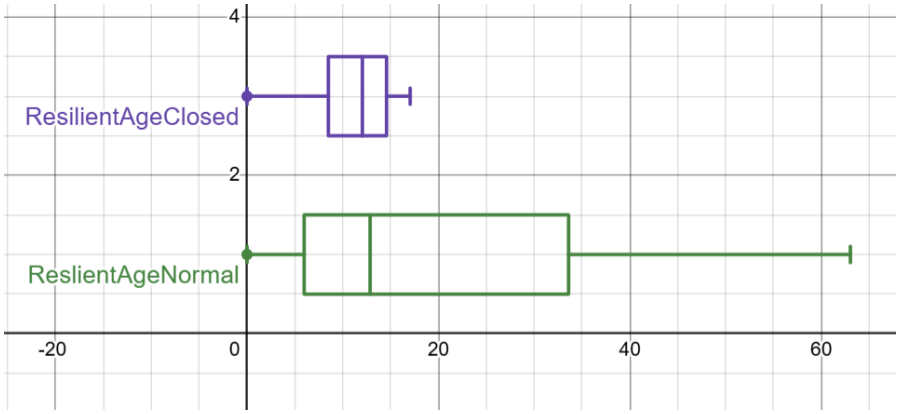
Discussion Question:
If you find a statistically significant difference in chlorine levels between the two zones, how can you prove it was the pipe closure that caused it, and not one of the variables listed above?

Teacher Notes and Graphs to Expect

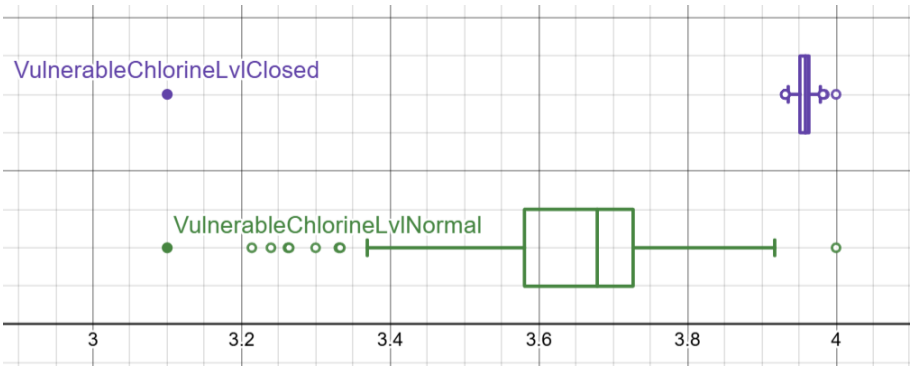
Water Age Vulnerable



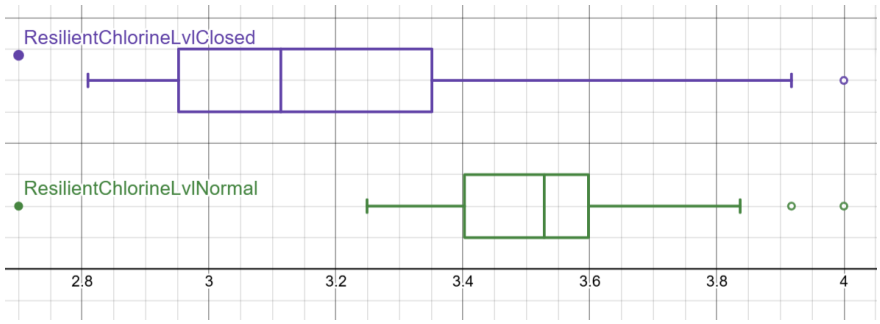
Water Age Resilient



Chlorine Level Vulnerable



Chlorine Level Resilient



Comments to students about what we are seeing and the need for more than a single sample:

Closing a pipe changes how the network routes water. That rerouting can sometimes bring fresher (more-chlorinated) water to places we'd expect to get worse, while other places get worse. Single random samples can show these surprising effects, which is why engineers look at system-wide measures and time-series, not a single reading

AP Statistics Curriculum Alignment

This page outlines the alignment between the "Impact of Pipe Closure on Water Quality" lesson set and the Learning Objectives (LO) and Essential Knowledge (EK) statements defined in the 2026-27 AP Statistics Course and Exam Description (CED).

Unit 1: Exploring One-Variable Data and Collecting Data

Topic 1.1: Introducing Statistics

LO 1.1.B: Determine an investigative question within a statistical study.

- **EK 1.1.B.2:** An investigative question should be posed so that the required data can be collected and analyzed.

LO 1.1.A: Identify components within a statistical study.

- **EK 1.1.A.4:** A population consists of all items or individuals of interest.

Topic 1.2: Variables

LO 1.2.B: Identify types of variables.

- **EK 1.2.B.2:** A quantitative variable takes on numerical values for a measured or counted quantity and generally has units of measure (e.g., Water Age in hours, Chlorine in mg/L).

Topic 1.6: Descriptions for One Quantitative Variable Distributions

LO 1.6.A: Describe distributions of quantitative one-variable graphical representations.

- **EK 1.6.A.1:** Descriptions include shape, center, and variability (spread) as well as any unusual features such as outliers, gaps, or clusters in context.

Topic 1.7 & 1.8: Summary Statistics and Boxplots

LO 1.7.A: Calculate measures of center and position for quantitative data.

LO 1.7.B: Calculate measures of variability for quantitative data.

LO 1.8.A: Construct quantitative one-variable graphical representations of summary statistics (Boxplots).

Topic 1.9: Comparisons of Distributions

LO 1.9.A: Compare multiple quantitative one-variable graphical representations. (Applied when comparing Vulnerable vs. Resilient zones).

Topic 1.10: The Investigative Question and Observational Studies

LO 1.10.A: Determine the components of an investigative question within a statistical study.

- **EK 1.10.A.1:** Component 1: Guide data collection (variables of interest).
- **EK 1.10.A.2:** Component 2: Guide data analysis (parameter and direction).
- **EK 1.10.A.3:** Component 3: Indicate type of conclusion and population.

LO 1.10.D: Identify an observational study.

- **EK 1.10.D.5:** A confounding variable provides an alternative explanation... it must be associated with both the explanatory and response variables.

Statistical Practices Addressed

Practice 1: Formulate Questions – Students determine valid investigative questions regarding infrastructure failure.

Practice 2: Collect Data – Students identify variables and justify the use of simulation data.

Practice 3: Analyze Data – Students construct boxplots/histograms and calculate summary statistics.

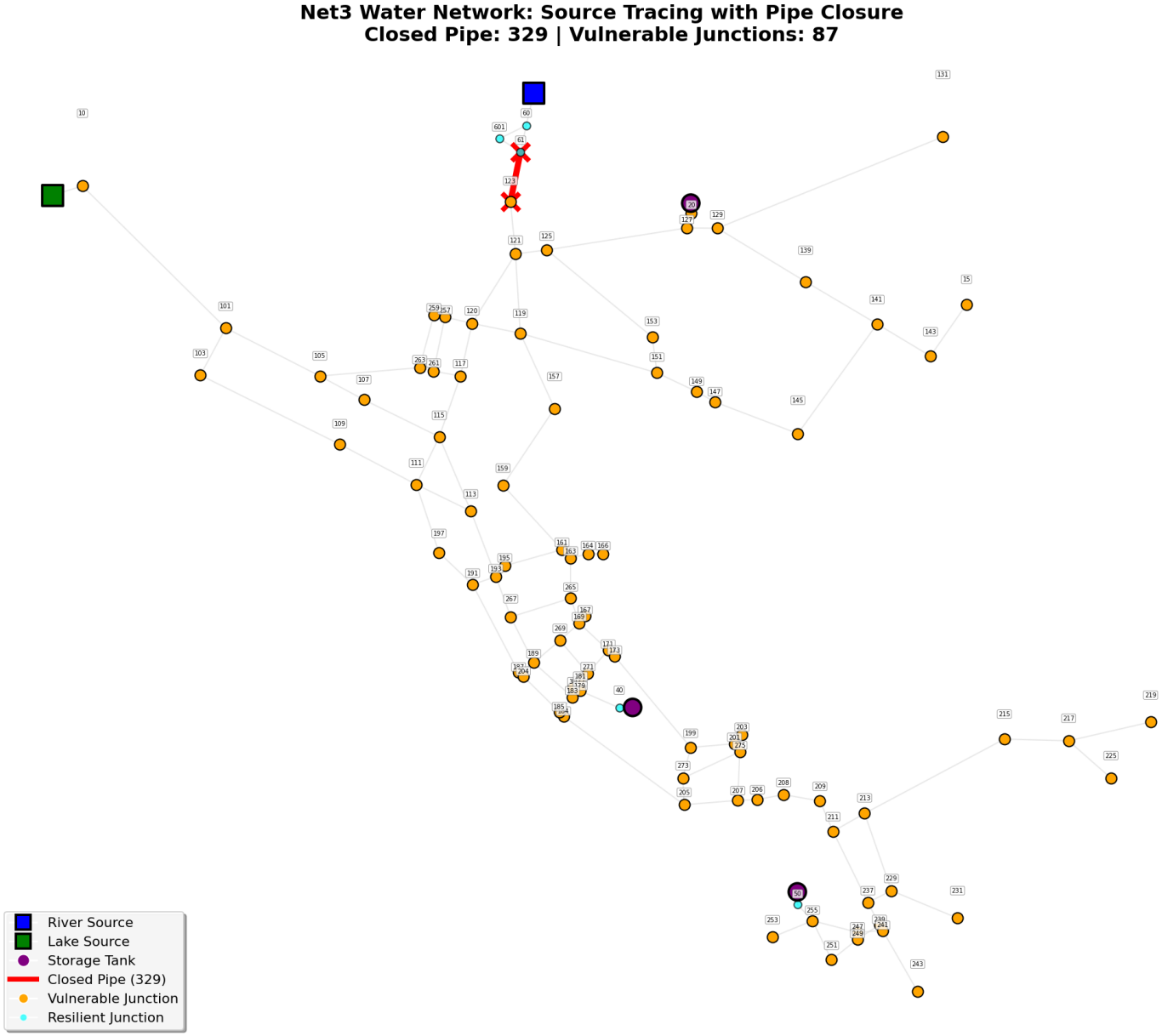
Practice 4: Interpret Results – Students describe distributions and justify claims about public health safety based on statistical evidence.

AP Statistic **Unit 1 - Infrastructure Failure & Source Mixing Analysis**
Sirois **Formative Assessment Writing Questions & Comparing Distributions**

Scenario: Investigating the Impact of Critical Pipe Failure on Dual-Source Water Distribution

Overview
In this investigation, you will act as a researcher at the UIC SWIM Lab. You are analyzing data from a water network simulation. Under normal conditions, the city draws water from both a **River** and a **Lake**. However, infrastructure is fragile. The SWIM Lab has simulated a "Critical Pipe Failure" by closing one of the top 10 most impactful pipes in the system (based on water pressure)

Your goal is to determine if this failure significantly alters the mixture of water (the percentage of Lake water vs. River water) reaching residents in different zones of the city. The data shows the averages of readings taken over the 3 days.



Formulating the Investigative Question

As strong statistical study begins with a well-stated question that guides the investigative process. As you covered in the previous lesson, a complete investigative question must include three specific components.

- **Component 1 (Data Collection):** Name the variable of interest and the groups being compared. Does your question help guide the data collection process?
- **Component 2 (Analysis):** Specify the parameter (e.g., mean) and the direction of the expected difference. Does your question guide how the data will be analyzed?
- **Component 3 (Conclusion):** State the population you are able to generalize your results to, and whether a cause-and-effect conclusion is possible. Does your question indicate what type of conclusions can be made from our study?

Study the following questions that were written by students setting up a statistical investigation.

Version A

How does the water mixture change after a critical pipe failure?

Version B

Does the critical pipe failure significantly alter the percentage of Lake water vs. River water reaching residents in different zones of the Net3 water network?

Version C

What is the average percentage of Lake water in the water mixture after a critical pipe failure, and how does this compare to the percentage before the failure?

Version D

Does the critical pipe failure significantly alter the percentage of Lake water vs. River water reaching residents in different zones of the Net3 water network, and can these findings be generalized to all junctions in the network?

Version E

Does the critical pipe failure significantly alter the water mixture in the city?

For the "formulating questions" assessment:

You will work in groups to:

1. Analyze each question version
2. Identify which component(s) are missing from each
3. Identify which version has all three components
4. Explain why the complete version is stronger

Part 2: Comparing Distributions

Open the dataset produced by the computational modeling and simulation tool and locate the **Lake Water Percentage** data for the **Vulnerable Zone** after the pipe closure. It can be found in the Excel file linked on the Topics page of On Campus. Work collaboratively to use Desmos to help with the analysis of the data in that file.

Write a paragraph comparing the distributions of Lake Water Percentage for the two zones. You must use comparative language (e.g., "greater than," "more skewed") and address the **four essential characteristics**:

- **Shape:** (Symmetry, skewness, peaks)
- **Outliers:** (Results from your 1.5 * IQR test)
- **Center:** (Compare means or medians)
- **Spread:** (Compare standard deviations or IQRs)

Part 3: "Histogram vs. Boxplot"
Compare the two types of displays for the Vulnerable Zone data.

What specific features of the distribution (e.g., gaps, clusters, multimodality) are visible in the **histogram** but are hidden in the **boxplot**?

Part 4: Formulating Insights & Conclusions
Based on your graphical analysis and numerical summaries, answer the following:

1. **Infrastructure Impact:** Did the pipe closure cause the Vulnerable Zone to rely more heavily on one specific source (River or Lake) compared to the Resilient Zone? Cite specific statistics to support your claim.
2. **Practical vs. Statistical Significance:** Suppose the mean Lake percentage in the Vulnerable Zone shifted from 45 percent to 48 percent. While this might be "statistically significant" in a large simulation, why might a City Health Official consider this shift to have low practical significance?

Teacher Notes

Sample Data Summary Normal Operation

Zone	River_Mean	River_Std	Lake_Mean	Lake_Std
Resilient	55.22999954	42.72999954	44.77000046	42.72999954
Vulnerable	41.59000015	34.06000137	58.40999985	34.06000137

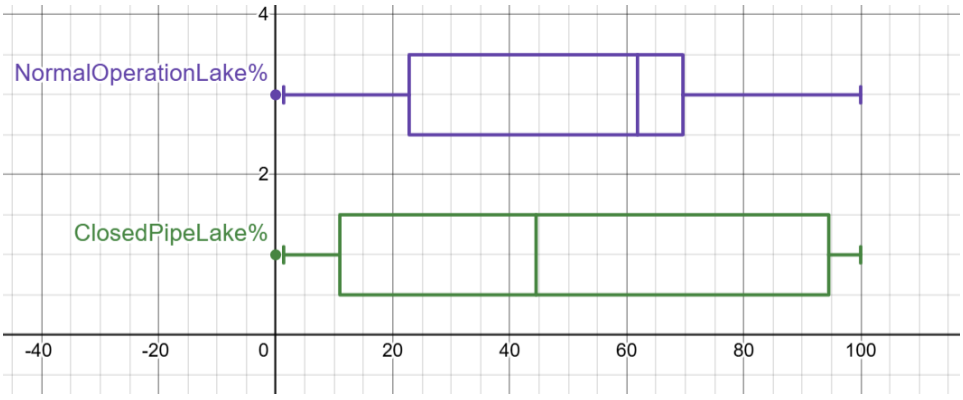
With Critical Pipe Closed

Zone	River_Mean	River_Std	Lake_Mean	Lake_Std
Resilient	68.58000183	41.79999924	31.42000008	41.79999924
Vulnerable	12.93000031	25.55999947	87.06999969	25.55999947

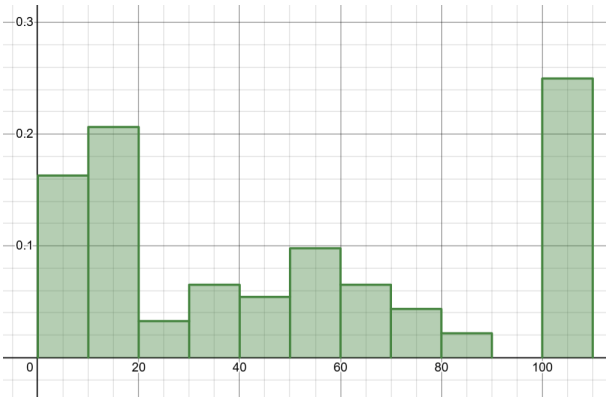
Metadata on above simulation

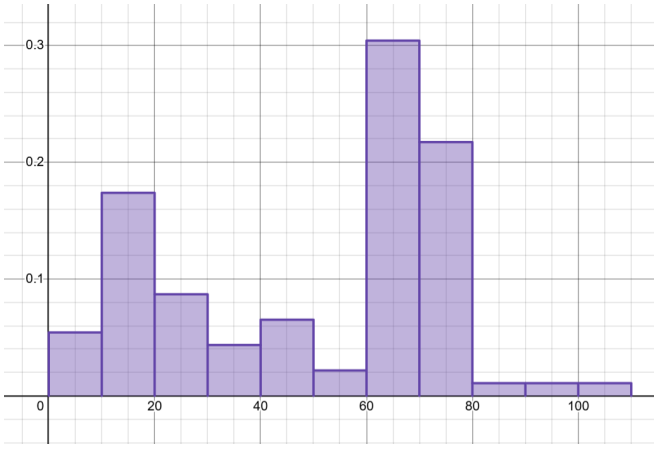
Parameter	Value
Closed Pipe	189
Pressure Drop Threshold	5.0 psi
Vulnerable Count	29
Resilient Count	63
Simulation Duration	72 hours
Total Junctions	92

Lake Water Mean Mixture Percentage Through Simulation



Opportunities for students to identify fences but no outliers and to identify what can features are lost from the data set when using a boxplot vs a dotplot or histogram





AP Statistics CED Alignment

Unit 1: Follow-up Activities for Formative Assessment Use

The following Learning Objectives (LO) and Essential Knowledge (EK) components from the 2026-27 AP Statistics Course and Exam Description are addressed in this formative assessment worksheet.

Section 1: Formulating Investigative Questions

- **LO 1.1.B:** Determine an investigative question within a statistical study.
- **EK 1.1.B.2:** An investigative question should be posed so that the required data can be collected and analyzed.
- **LO 1.10.A:** Determine the components of an investigative question within a statistical study.
- **EK 1.10.A.1:** The first component of an investigative question should guide the data collection process and should be phrased in terms of the variable(s) of interest in the study.
- **EK 1.10.A.2:** The second component of an investigative question should guide the data analysis choice.
- **EK 1.10.A.3:** The third component of an investigative question should indicate the type(s) of conclusion(s) applicable from the study. The investigative question should provide the population to which the conclusions will be applicable.

Section 2: Describing and Comparing Distributions

- **LO 1.6.A:** Describe distributions of quantitative one-variable graphical representations.
- **EK 1.6.A.1:** Descriptions of the distribution of one quantitative variable include shape, center, and variability (spread) as well as any unusual features such as outliers, gaps, or clusters in context.
- **LO 1.7.D:** Calculate outliers for quantitative data.
- **EK 1.7.D.1.i:** An outlier is a value located more than 1.5×IQR above the third quartile or more than 1.5×IQR below the first quartile.
- **LO 1.9.A:** Compare multiple quantitative one-variable graphical representations.
- **LO 1.9.B:** Compare multiple quantitative one-variable graphical representations of summary statistics.
- **LO 1.9.C:** Justify a claim using multiple quantitative one-variable graphical representations.

Section 3: Graphical Displays (Histograms vs. Boxplots)

- **LO 1.5.A:** Construct quantitative one-variable graphical representations.
- **EK 1.5.A.2:** A histogram places the observed values of the quantitative variable into ordered intervals... Altering the interval widths can change the appearance of the histogram.
- **LO 1.8.A:** Construct quantitative one-variable graphical representations of summary statistics (Boxplots).
- **EK 1.8.A.2:** A boxplot is a graphical representation of the five-number summary... If there are outliers in the data, the whiskers extend to the most extreme data values that are not outliers.
- **LO 1.9.A:** Compare multiple quantitative one-variable graphical representations. (Specifically: identifying features like gaps, clusters, or multimodality visible in histograms but hidden in boxplots).

Section 4: Statistical vs. Practical Significance

- **LO 4.5.A:** Carrying out a test for a population mean or population mean difference.
- **EK 4.5.A.4:** Statistical significance means that the observed results are unlikely to occur by chance alone. Practical significance means that the observed results are large enough to be important in the real world.

Section 5: Generalization and Scope of Inference

- **LO 1.10.E:** Justify the appropriateness of generalizations for a statistical study.
- **EK 1.10.E.2:** When observational units, or experimental units, in a sample are randomly selected from a population, it is appropriate to make generalizations about the entire population of individuals from which the sample was selected.

AP Statistics **Multi-Event Infrastructure Analysis & Sensor Optimization**
Sirois **Using the Mean and Standard Deviation in Data Analysis**

Scenario: 7-Day System Stress Test (Fire Flows & Pump Malfunction)
WNTR Code: Applications of Standard Deviation in Data Analysis - Multi-Event Sensor Questions.py

The "Perfect Storm" Scenario
As a data analyst for the city's water utility, you are monitoring the Net3 water network over a 168-hour (7-day) period. During this week, the system was subjected to three major stressors:

1. **Day 3 (Hours 48–54):** A major industrial fire requiring high-volume "Fire Flow."
2. **Day 5 (Hours 96–108):** The primary pump (Pump 1) was shut down for emergency maintenance.
3. **Day 6 (Hours 132–138):** A secondary residential fire occurred while the system was still recovering from the pump shutdown.

Your objective is to use variability analysis to determine where the city should invest in permanent monitoring sensors.

Analysis #1: Spatial Context & Predictions

Refer to the **Infrastructure Impact Map** provided by the simulation to get a big picture overview of how the water infrastructure is laid out.

1. **Identify the Stressors:** Locate the two fire flow junctions (marked with stars) and the primary pump (marked with a warning symbol).
2. **Initial Hypothesis:** Which junctions do you predict will show the highest **standard deviation** in pressure? Which will show the highest mean water age? Explain your reasoning.

Analysis #2: Variability as an Information Metric
Open the **Perfect_Storm_Data.xlsx** file. You have hourly data for **Water Age**, **Chlorine**, and **Pressure** for all junctions. The data contains both the mean and standard deviation, as well as maximum and minimum data for each junction during the 7-day (168-hour) simulation.

High variability relative to the mean may be an indication of the need for monitoring of a junction.

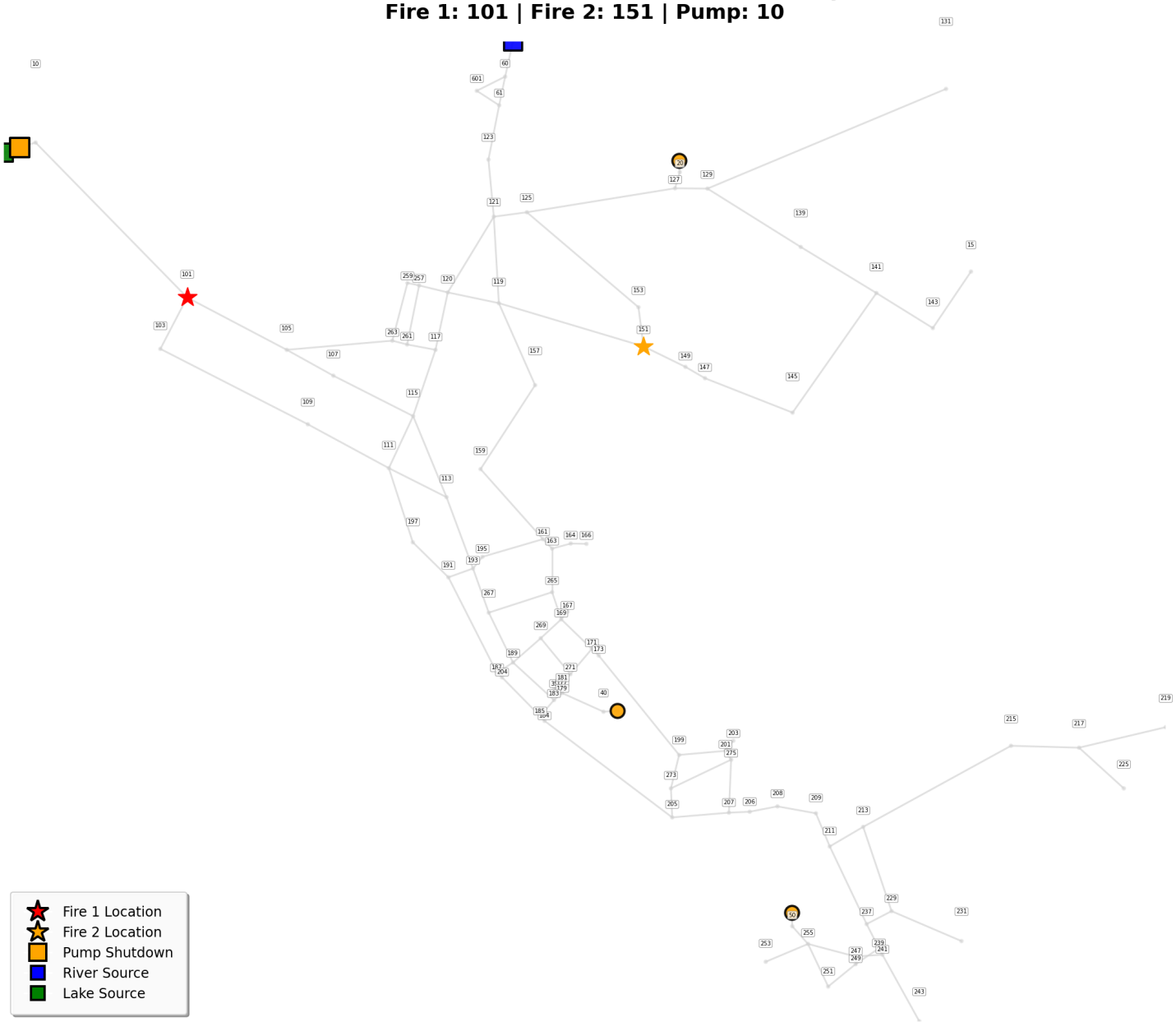
Chronic Problems vs. Acute Failures
Statisticians distinguish between **chronic** (consistent) and **acute** (variable) issues.

1. **Chronic Problem Areas:** Identify junctions where the **Mean Water Age** is >40 hours but the **Standard Deviation** is low (<5 hours). What does this tell you about the water flow in this area?
2. **Acute Failure Indicators:** Identify junctions where the **Standard Deviation of Pressure** is in the top 5% of the network. Are these junctions located near the fire events?

Situations that may be worth monitoring with a sensor:

- Normal pressure with high variability (may be triggered by an event, and the sensor would notify you of an issue to investigate). Why is a junction with high variability but a low average a potentially "informative" site?
- Low chlorine levels and low variability (consistently bad water quality)
- High age levels and low variability (older water tends to be stagnant and the chlorine levels tend to decay over time)
- High variability for either chlorine or the age of the water may be indicative of an issue.

Perfect Storm: Multi-Event Infrastructure Analysis
Fire 1: 101 | Fire 2: 151 | Pump: 10



Analysis #3: Event-Specific Time-Series Analysis – Do this when viewing the time-series analysis after you've identified junctions you think are candidates for sensor placement

The Pump Shutdown (Hours 96–108)
Look at the pressure data during the 12-hour pump outage.

- Compare the **Mean Pressure** during the outage to the **Baseline Mean** (Hours 0–47).
- **Statistical Insight:** Did the system return to its baseline mean immediately after the pump was restarted at Hour 108? Use the data to describe the "recovery curve."

Compound Effects (The Day 6 Fire)
The second fire occurred at Hour 132.

- Compare the **minimum pressure** reached during Fire 1 (Normal Pump) vs. Fire 2 (Post-Pump Failure).
- **Argument:** Does the data suggest that a system is more "vulnerable" to secondary events while recovering from a primary failure?

Decision Time: Sensor Placement Recommendation
You have a budget for **4 permanent sensors**.
Task: Based on your analysis of means and standard deviations, propose a placement strategy.

- How many sensors will you place at **High-Variability** sites (for early warning of fires/breaks)?
- How many sensors will you place at **High-Mean/Low-Standard Deviation** sites (to monitor chronic water quality issues)?

- **Justification:** Pick one specific Junction ID and justify its sensor placement using both a measure of center and a measure of spread.

Statistical vs. Practical Significance
A junction 2 miles away from the fire showed a pressure drop from 60 psi to 58 psi. The p-value for this change is <0.01, making it statistically significant.

- **Critical Thinking:** Does this change have **practical significance** for the residents? Would you prioritize a sensor at this location? Why or why not?
- A different junction shows a chlorine drop from 0.3 mg/L to 0.1 mg/L to be both statistically and practically significant. Explain what this means, in context of the water infrastructure scenario.

Teacher Activity Guide: Multi-Event Infrastructure Simulation & Sensor Optimization

1. Instructional Overview

This activity uses a computational modeling and simulation (CMS) approach to teach core Unit 1 and Unit 2 concepts. Students act as data analysts for a water utility, using measures of center and spread to optimize the placement of 5 sensors across a complex network (Net3) subjected to multiple stressors (fire flows and pump failures).

Statistical Learning Objectives

- **Measures of Center (mean):** Identifying chronic risk areas.
- **Measures of Spread (standard deviation):** Identifying informative, early-warning sites.
- **Time-Series Context:** Understanding how distributions change over a period of time (determined by the length we run the simulation, currently 1 week)
- **Statistical vs. Practical Significance:** Making engineering decisions based on data magnitude (possibly, if we use this to

2. Phase I: Initial Simulation & Data Generation

Before class, run the WNTR "Perfect Storm" simulation script. This script models 168 hours (7 days) of system activity. It creates an excel file that contains pressure, chlorine and water age readings.

Step-by-Step Workflow:

1. **Run the Script:** Execute the Python code to generate Perfect_Storm_Data.xlsx. Modify this data set to provide students with 1 sheet of data they can use to copy and paste into Desmos or just to visually analyze.
2. **Verify the Output:** Ensure the Excel file contains hourly rows for all 92 junctions with columns for Pressure_PSI, Chlorine_mgL, and Water_Age_Hrs.
3. **Distribute Materials:** Provide students with the Excel file, the **Infrastructure Impact Map**.
4. **Student analysis.** Students should collaboratively make decisions on potential sites for sensor placement. Look in the code for "Selected_Junctions" to modify the list to include their choices. Re-run the simulation to produce time-series analysis that can lead to a discussion on the relationship between the 3 variables, as well as whether the selected junction would be a good candidate for placing a sensor.

3. Phase II: Guiding Student Analysis

Students should use Excel filtering or Desmos to identify two distinct types of "prime candidates" for sensors. Use this as a time to discuss the periodic nature of water demand and therefore, the periodic nature of the time-series graphs.

Strategy A: The "Early Warning" Site (High Variability)

- **Variable focus:** Pressure.
- **Statistical Profile:** Normal mean, but extremely high standard deviation.
- **Logic:** These junctions are potentially more sensitive to system shocks. When a pipe breaks or a fire starts, these locations show may be the ones that indicate that an event has happened before other sites.
- **Pro Tip:** Encourage students to look for junctions near the fire stars on the map, but also "bottleneck" junctions that serve large areas. Discuss loop areas or end nodes and how they may differ from other junctions.

Strategy B: The "Chronic Problem" Site (Extreme Means with Low Standard Deviation)

- **Variable focus:** Water Age or Chlorine.
- **Statistical Profile:** High mean age or low mean chlorine with low standard deviation.
- **Logic:** These are "consistently poor" quality areas. If a junction always has 100-hour-old water and 0.1 mg/L chlorine, it is a public health risk that may require more consistent monitoring, even if it doesn't "react" to specific events.
- **Teacher Tip:** Point students toward "end nodes" or dead-ends (e.g., Junctions 243, 40, 20 are some to consider) where water stagnates.

4. Phase III: Validation via Time-Series Re-Run

Once student groups have identified their top 3 junction candidates, re-run the validation plotting script for those specific IDs that can lead to a full class discussion of what seems to be going on within the system. Use the following to possibly guide student collaboration:

- Which junctions showed the most surprising behavior?
- Did any junctions challenge our initial hypotheses about network topology?
- How did the time-series graphs confirm or contradict our statistical conclusions?

The Validation Process:

1. **Input Junctions:** Enter the student-selected IDs into the selected_junctions list in the plotting script.
2. **Generate 3x3 Grid:** Produce the time-series plots showing Pressure, Chlorine, and Age for those three sites.
3. **Analyze the "Why":**
    - **Linear Increases:** If Age increases linearly ($y = x$), the junction is a simple pass-through (e.g., Junction 601). Is this a good sensor site? (Usually no, it's too predictable).
    - **Dramatic Spikes:** If Age spikes from 10 to 110 hours (e.g., Junction 253), it indicates a complex recovery or tank-drainage event. Is this a good sensor site? (Yes, it's highly informative that something has taken place).
    - **Flatlines at Danger Levels:** If Chlorine stays below 0.2 mg/L throughout (e.g., Junction 243), it confirms a chronic failure.

5. Connecting to AP Statistics Concepts

Standard Deviation as "Information"

Provide students with opportunities to explain system changes in context, using standard deviation as a talking point. Connect the measure with what is displayed on the graph. Help students move beyond seeing s as just a "spread" number. In this context, s is a measure of sensitivity to events that are occurring in the infrastructure. A junction with $s \approx 0$ tells us nothing new about the system's state, while a high s provides a high "Value of Information."

Statistical vs. Practical Significance (Not sure if we will work with this during this activity or not… Don't overload too many concepts here)

We could use the pressure data to drive this home.

- **Scenario:** A fire causes an average pressure drop from 60 psi to 57 psi. With n=168, this may be statistically significant but may be imperceptible to the community.

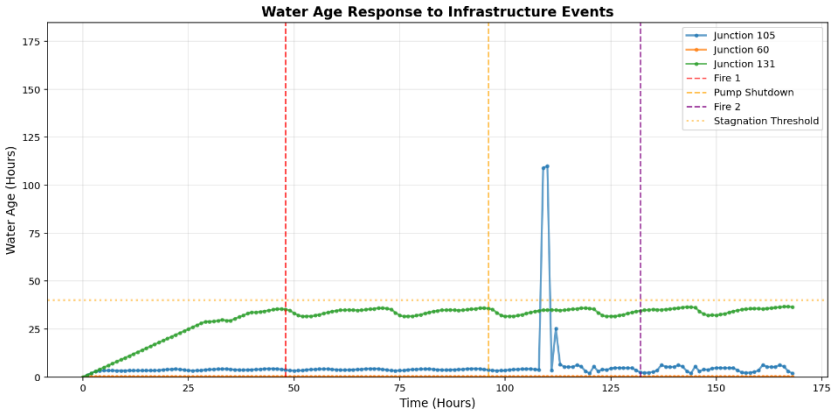
- **Question:** Is the pressure change practically significant?
- **Answer:** No. Residents won't notice a 3 psi drop. However, a Chlorine drop from 0.3 mg/L to 0.1 mg/L is both statistically and practically significant (it crosses the safety threshold).

More Teacher Notes on Data and Graphs Produced

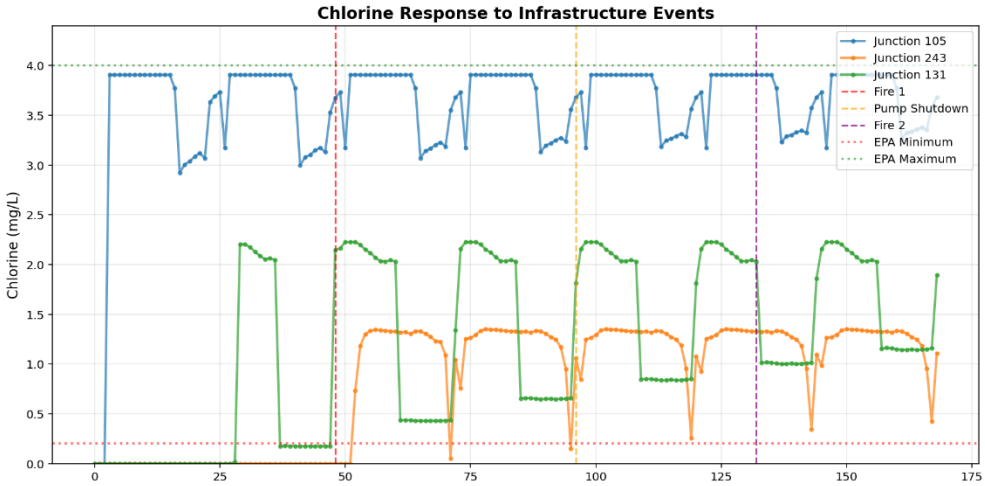
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Junction	Pressure_Mean	Pressure_Std	Pressure_Min	Pressure_Max	Chlorine_Mean	Chlorine_Std	Chlorine_Min	Chlorine_Max	Age_Mean	Age_Std	Age_Min	Age_Max
2	10	17.13999939	13.92000008	-0.660000026	29.65999985	3.779999971	0.479999989	0	4	34.83000183	36.25	0	108
3	101	38.20999908	4.630000114	31.35000038	44.40000153	3.640000105	0.519999981	0	3.930000067	4.420000076	8.170000076	0	109
4	103	37.47999954	4.28000021	31.04000092	43.43000031	3.559999943	0.610000014	0	3.920000076	6.119999886	11.47000027	0	109.7799988
5	105	40.50999832	2.849999905	35.88999939	45.22999954	3.599999905	0.579999983	0	3.910000086	5.28000021	11.60000038	0	110
6	107	42.36000061	2.75	37.86999893	47	3.559999943	0.600000024	0	3.880000114	7.389999866	12.43999958	0	110.0599976
7	109	43.36999893	3.410000086	37.95999908	48.61000061	3.589999914	0.560000002	0	3.880000114	7.690000057	11.68000031	0	110
8	111	45.02999878	2.059999943	41.49000168	48.88000107	3.559999943	0.550000012	0	3.859999895	4.940000057	7.559999943	0	74.02999878
9	113	47.09000015	1.74000001	43.93999863	50.52000046	3.50999999	0.560000002	0	3.829999924	5.650000095	5.980000019	0	60.20999908
10	115	43.93000031	2	40.47000122	47.77000046	3.549999952	0.560000002	0	3.839999914	4.159999847	3.150000095	0	31.86000061
11	117	44.40999985	1.74000001	41.45999908	48.00999832	3.599999905	0.550000012	0	3.869999886	3.339999914	0.639999986	0	4.840000153

Have students identify junctions that might be indicators of high standard deviation relative to the mean AND/OR "Poor Readings" with low variability.

Identify 3 junctions and then run the simulation again, to check the 3 time-series graphs produced. This will be their introduction to time-series analysis. Go slow with this.



This time-series indicates a spike at Junction 105 which a sensor would detect. It follows the pump shutdown.



This time-series plot identifies junction 243 as a potential location for a sensor because of low chlorine readings and low variability. This is especially true after the 1st fire.

Iteratively re-run the simulation taking their input and identifying if the locations would be appropriate based on the time-series analysis.

The "Lazy River" Concept (Included here for the teacher. I had a tough time understanding this when I was analyzing the data. This is the best I could come to explaining it for "students" if those are junctions they select to place sensors at. I want it documented for potential later reference.)

This refers to the phenomenon witnessed when viewing the time-series plots for junctions 60, 61, and 601. These form a triangle on the map.
My understanding is:

Why 60 and 61 Have "Fresher" Water

Junction 60:

Gets water directly from the source (or close to it)
This water hasn't traveled far yet
Age = low (fresher)

Junction 61:

Gets water that's traveled further from the source
But it also has OTHER paths bringing fresh water to it
Age = moderate (fresher than 601, but older than 60)

Junction 601:

Only gets water from 60 (one source)
No other paths bringing fresh water
Age = highest (oldest of the three)

Other regions of the infrastructure behave in a similar manner

If students notice that, over time, Age increases in a perfect linear manner, while Chlorine stays periodic, use the "Lazy River" analogy:

- **Age** is an odometer (accumulation).
- **Chlorine** is a fuel gauge (decay), but it is masked by the "refilling" of the river through daily demand cycles.
- This explains why the correlation between Age and Chlorine might be weaker than expected in a dynamic system.

6. Classroom Discussion Prompts

1. "If you only have money for ONE sensor, do you put it where the water is always bad, or where the water changes the most?"
2. "Why did Junction 253 take 50 hours to recover while Junction 10 recovered in 2 hours? How does the network map explain this?"
3. "Does a high standard deviation always mean the system is failing? Or could it just mean the system is responding correctly to a stressor?"

AP Statistics CED Alignment

Unit 2: Data Analysis with Standard Deviation Focus (Multi-Event Sensor Placement)

The following Learning Objectives (LO) and Essential Knowledge (EK) components from the 2026-27 AP Statistics Course and Exam Description are addressed in this infrastructure analysis and sensor optimization activity.

Section 1: Describing and Comparing Distributions

- **LO 1.6.A:** Describe distributions of quantitative one-variable graphical representations.
- **EK 1.6.A.1:** Descriptions of the distribution of one quantitative variable include shape, center, and variability (spread) as well as any unusual features such as outliers, gaps, or clusters in context.
- **LO 1.7.A:** Calculate measures of center and position for quantitative data.
- **LO 1.7.B:** Calculate measures of variability for quantitative data.
- **EK 1.7.B.4:** The standard deviation is a typical deviation of the data values from their mean... The sample standard deviation is denoted by s.
- **LO 1.7.F:** Justify the selection of a summary statistic for describing quantitative data.
- **EK 1.7.F.1:** The median and IQR are considered a resistant (or robust) measure of center and measure of variability... the mean is considered a nonresistant measure of center, and the range and standard deviation are considered nonresistant measures of variability.
- **LO 1.9.A:** Compare multiple quantitative one-variable graphical representations.

Section 2: Relative Position and Standardized Scores

- **LO 1.9.D:** Calculate z-scores with population parameters.
- **EK 1.9.D.1:** A standardized score measures the number of standard deviations a data value falls above or below the mean.
- **LO 1.9.E:** Compare z-scores as measures of relative position for distributions.

Section 3: Normal Distributions and Empirical Rule

- **LO 2.11.A:** Determine probabilities for a normal distribution.
- **EK 2.11.A.2:** For a normal distribution, the Empirical Rule states that approximately 68% of the area... is within 1 standard deviation of the mean, approximately 95%... within 2 standard deviations, and approximately 99.7%... within 3 standard deviations. (Relevant to identifying "top 5%" or "acute failure" indicators).

Section 4: Statistical vs. Practical Significance

- **LO 4.5.A:** Carrying out a test for a population mean or population mean difference.
- **EK 4.5.A.4:** Statistical significance means that the observed results are unlikely to occur by chance alone. Practical significance means that the observed results are large enough to be important in the real world.

Section 5: Regression and Time-Series Context (Unit 5 Preview)

- **LO 5.1.A:** Construct graphical representations between two quantitative variables (Scatterplots/Time-Series).
- **LO 5.2.A:** Calculate and interpret the correlation coefficient (r).
- **EK 5.2.A.3:** Correlation does not imply causation. (Relevant to the "Lazy River" concept and the relationship between Water Age and Chlorine).

Section 6: Formulating Investigative Questions

- **LO 1.1.B:** Determine an investigative question within a statistical study.
- **LO 1.10.A:** Determine the components of an investigative question within a statistical study.
- **EK 1.10.A.1:** The first component... should guide the data collection process.
- **EK 1.10.A.2:** The second component... should guide the data analysis choice.

AP Statistics **Water Quality & Infrastructure Analysis**
Unit 2: Probability, Random Variables, and Probability Distributions

Review of Background Information: The Physics of the System
In a water distribution network, chlorine is added at the reservoir to kill bacteria. As water travels through pipes, the chlorine naturally decays. The EPA recommends a chlorine residual between **0.2 mg/L and 5.0 mg/L** for safety. When a "Critical Pipe" fails, water is forced to take circuitous detours, increasing the **Water Age** (travel time) and potentially allowing chlorine to drop below safe levels (chlorine in the system decays over time).

Summary of Variable Relationships

Variable	Type	Basis in Simulation	Criteria
Zone Type	Categorical	Hydraulic Pressure (psi)	Vulnerable (psi drop of 5+) Resilient (psi drop < 5)
Safety Status	Categorical	Chlorine Concentration (mg/L)	Safe 0.2 <= chlorine <= 5 mg/L Unsafe if out of that range
Water Age	Quantitative	Travel time from Reservoir (hrs)	
Chlorine Level	Quantitative	Chemical Decay over time	

Physical Context to Keep in Mind
Throughout these lessons, while our analysis will focus on the "math" of the association, keep in mind that the reason for the association is physical:
1. Pipe Break → Pressure Drop in Nearby Junctions → Detour of water in the system around the break.
2. Detour → Increased Water Age (because it may take a longer path through the network)
3. Increased Water Age → Chlorine Decay (there is a bulk decay rate variable used in the model)
4. Chlorine Decay → Unsafe Status (if enough loss occurs)

Part 1: Building the Contingency Table
You have been provided with data that resulted from 100 independent simulations of water infrastructure that has experienced a critical pipe failure. In each simulation, one junction was randomly selected from the **Vulnerable Zone** (the pressure reading for that junction indicated that it was directly impacted by the pipe failure) and one junction was randomly selected from the **Resilient Zone** (the pressure readings for that junction indicate that it was not impacted by the pipe failure). Depending on which critical pipe failed, there may not be any vulnerable junctions. These readings were taken 24 hours after the pipe closing.

Task 1: Use the provided dataset (Water_Safety_Phase1_Closed.xlsx) to complete the 2×2 contingency table below.

Zone Type	Safe (0.2 – 5.0 mg/L)	Unsafe (< 0.2 mg/L)	Total
Resilient			
Vulnerable			
Total			

Task 2: Probability Analysis
Using your table from Part 1, calculate the following probabilities. Round to four decimal places.

1. **Marginal Probability:** What is the probability that a randomly selected observation from this study is classified as "Unsafe"? Use proper notation.

2. **Conditional Probability:** Given that a junction is in the **Vulnerable Zone**, what is the probability that the water is "Unsafe"?

3. **Conditional Probability:** Given that a junction is in the **Resilient Zone**, what is the probability that the water is "Unsafe"?

Task 3: Evaluating Independence

Two events are independent if the occurrence of one does not change the probability of the other. Mathematically, A and B are independent if $P(A) = P(A|B)$.

1. Before beginning with a probability analysis, discuss if you believe the safety of water depends on the zone it comes from, in the context of this scenario.

2. Compare your results for P(Unsafe) and P(Unsafe|Vulnerable). Are they equal?

3. Based on this comparison, are the variables **Zone Type** and **Safety Status** independent? Justify your answer using the specific probabilities you just calculated.

4. Find the probability that a randomly selected junction will have unsafe water when a critical pipe is closed.

5. From your contingency table, find $P(\text{Unsafe} \cap \text{Vulnerable})$, P(Unsafe or Vulnerable)

Task 4: Analyzing Data after the Pipe was Re-opened
Now, open the Excel file Water_Safety_Phase2_Reopened.xlsx and build a new contingency table. The data in this file was obtained by opening the closed pipe, and then running the simulation for another 2 hours to help determine if that is enough time to get the system back to normal.

Zone Type	Safe (0.2 – 5.0 mg/L)	Unsafe (< 0.2 mg/L)	Total
Resilient			
Vulnerable			
Total			

1. Find the probability that a randomly selected junction will have unsafe water 2 hours after the closed pipe was opened.

2. Consider the contingency table that you built from the simulation run 2 hours after the closed pipe was opened. Based solely on that table, are the events "unsafe chlorine levels" and "junction in the vulnerable zone" mutually exclusive events? What is the probability in this simulation, that water is both unsafe and from a vulnerable zone? Explain why this makes sense in the context of the recovery process. Why might this not be true for the resilient zone?

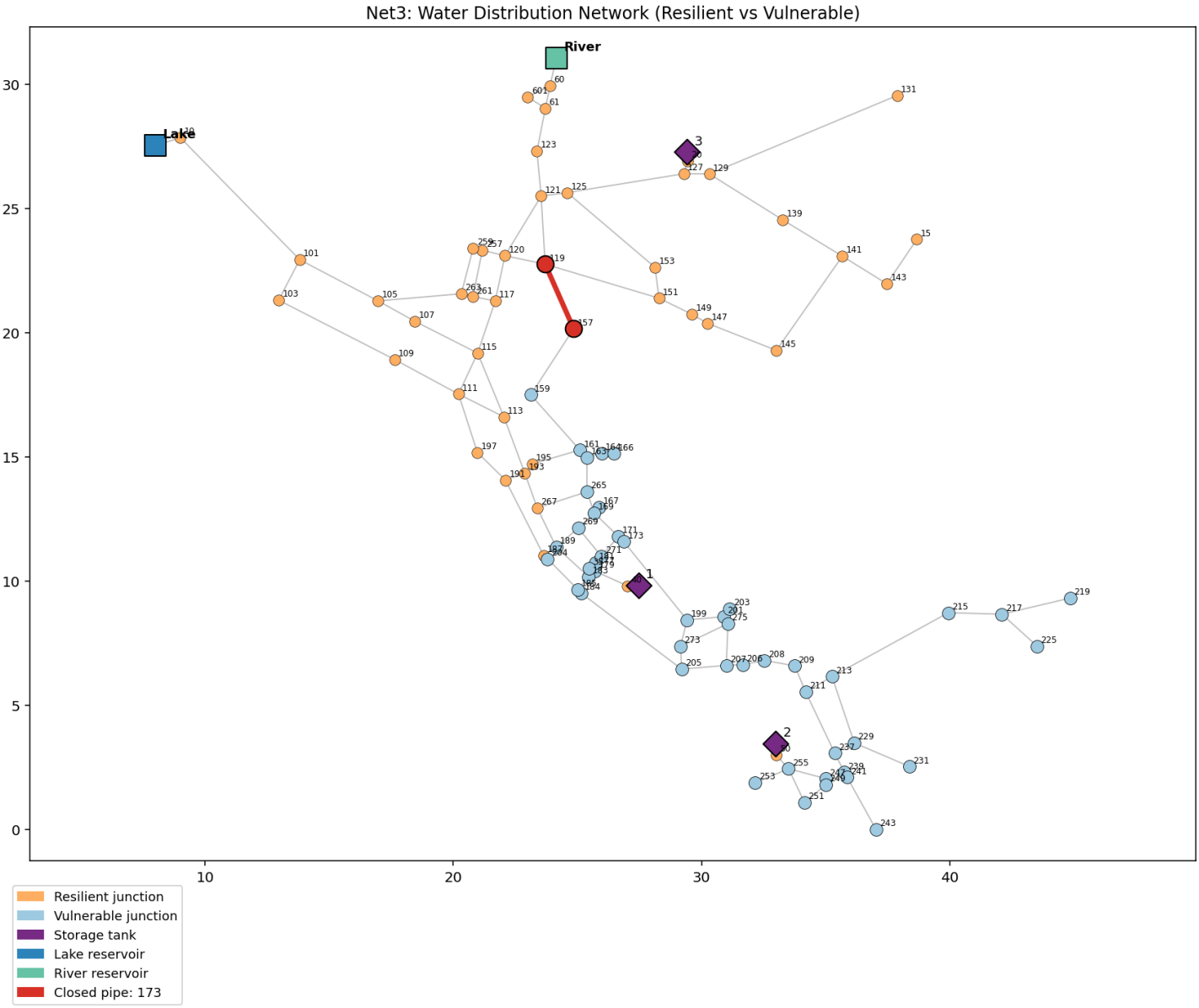
3. **Contextual Reasoning:** Compare the relationship between Zone Type and Safety Status in Task 3 (pipe closed) versus Task 4 (pipe reopened). Did the relationship change? Explain the physical reason for this change.

4. **Another contextual observation:** The zones that were most impacted during the failure are sometimes the first to benefit when the failure is resolved. Does this appear to be the case in the scenario you just analyzed? Support your answer with specific evidence from your contingency table

Task 5: Simulation and the Law of Large Numbers
In this study, we ran 100 trials.

1. If we increased the simulation to **10,000 trials**, what would you expect to happen to the value of the empirical probability P(Unsafe|Vulnerable)?

2. Explain, in context, why is a 10,000-trial simulation more useful for a City Health Official than a 100-trial simulation?


Net3: Water Distribution Network (Resilient vs Vulnerable)
River
Lake
1
2
3
Resilient junction
Vulnerable junction
Storage tank
Lake reservoir
River reservoir
Closed pipe: 173

Teacher Notes

The simulation will produce an excel file for students to use to get used to how contingency tables are built. Don't require that they build the entire thing, but use it to make connections. A snapshot of part of a sample sheet from a closed pipe simulation is shown below. Notice that some trials have 0 vulnerable. On each simulated trial, 1 randomly selected junction from a resilient zone and 1 from a vulnerable zone is displayed EXCEPT in the case where none are vulnerable. Dispel the myth that students need to have equal sample sizes. Start with the idea that we want both resilient zones and vulnerable zones represented. Indicate that this is the idea behind stratifying.

	A	B	C	D	E	F	G	H	I
1	Trial_ID	Failed_Pipe	Zone_Type	Chlorine_mgL	Age_Hrs	Safety_Status	Num_Resilient	Num_Vulnerable	
2	1	50	Resilient	1.903872848	2.364171982	Safe	92	0	
3	2	179	Resilient	1.997597694	0.057656202	Safe	52	40	
4	2	179	Vulnerable	1.526894689	6.560752869	Safe	52	40	
5	3	20	Resilient	1.734464765	6.847139835	Safe	92	0	
6	4	40	Resilient	1.816386223	4.619141102	Safe	92	0	
7	5	40	Resilient	1.735217214	6.821716785	Safe	92	0	
8	6	123	Resilient	3.02746E-08	11.99984741	Unsafe	57	35	
9	6	123	Vulnerable	1.843303561	3.913761377	Safe	57	35	
10	7	40	Resilient	1.778740644	5.624709129	Safe	92	0	
11	8	175	Resilient	1.794372559	5.203590393	Safe	44	48	
12	8	175	Vulnerable	0	12	Unsafe	44	48	
13	9	175	Resilient	1.80822897	4.834338188	Safe	44	48	
14	9	175	Vulnerable	0	12	Unsafe	44	48	
15	10	175	Resilient	1.8029778	4.974303246	Safe	44	48	
16	10	175	Vulnerable	0	12	Unsafe	44	48	
17	11	20	Resilient	0	12	Unsafe	92	0	
18	12	175	Resilient	1.802690744	4.981773853	Safe	44	48	
19	12	175	Vulnerable	1.425328493	7.27461195	Safe	44	48	
20	13	183	Resilient	1.680156589	8.357983589	Safe	75	17	
21	13	183	Vulnerable	1.622717142	6.214669704	Safe	75	17	
22	14	123	Resilient	1.745765686	6.520236969	Safe	57	35	
23	14	123	Vulnerable	1.778318167	5.651204109	Safe	57	35	

Closed Pipe will almost always result in dependent events as shown below:

```
--- Contingency Table ---
Safety_Status  Safe  Unsafe  All
Zone_Type
Resilient        83      17  100
Vulnerable       40      33   73
All             123      50  173

Marginal Probability P(Unsafe): 0.2890
Conditional Probability P(Unsafe | Vulnerable): 0.4521

--- Independence Evaluation ---
0.2890 ≠ 0.4521. Variables appear Associated.
```

Reopened will usually result in independent:

```
--- Contingency Table ---
Safety_Status  Safe  Unsafe  All
Zone_Type
Resilient        95       5  100
Vulnerable       71       6   77
All             166      11  177

Marginal Probability P(Unsafe): 0.0621
Conditional Probability P(Unsafe | Vulnerable): 0.0779

--- Independence Evaluation ---
0.0621 ≈ 0.0779. Variables appear Independent.
Script complete
```

Be sure to start discussing criteria for independence that can lead into Chi Square later…

AP Statistics **Unit 2 Part 2**
Sirois **Discrete Random Variables & Binomial Distributions**

Review of Background Information
In our previous lessons, we analyzed the impact of a "Critical Pipe" failure on a water distribution network. We discovered that when a pipe closes, water age increases, leading to chlorine decay. We calculated both marginal and conditional probabilities of a junction being "Unsafe" across the entire network during a failure. These were empirical calculations since they were based off of results of a simulation. We'll use the same approach in this investigation.

In this activity, we move from looking at individual junctions to looking at **counts** of unsafe junctions in a sample. This allows us to quantify the system-wide risk of a failure.

Part 1: The Binomial Model (BINS)
Before we analyze our new simulation data, we must determine if a Binomial Distribution is an appropriate model for our sampling process. For a random variable X to follow a binomial distribution, it must satisfy the following conditions (we'll use the acronym BINS):

- **B – Binary?** Are there exactly two possible outcomes for each trial?
- **I – Independent?** Does the outcome of one trial provide no information about the outcome of another?
- **N – Number?** Is there a fixed number of trials (n) set in advance?
- **S – Success?** Is the probability of success (p) the same for each trial?

Part 2: Empirical Data Collection
We will run a simulation involving a fixed sample size and a fixed number of randomized trials, recording the number of **unsafe** junctions. Notice that in this activity, P(Success) = P(Unsafe). Success doesn't necessarily mean a positive outcome when working with binomial probability, it means probability of obtaining an outcome we were looking for.

Keep in mind that in each trial, a critical pipe was closed, and a random sample of junctions was taken from across the entire network. The random sample was taken selecting junctions without replacement.

Task 1: Build a complete probability distribution of *X* for the simulation that was run. Write it in table form.

Task 2: Calculate the **mean and standard deviation** of your empirical data. Interpret each of those values in the context of the problem we are studying.

Part 3: Theoretical Binomial Probability

Now, let's compare your experimental results to the theoretical Binomial Distribution. Use the value of P(Unsafe) that we calculated in our previous lesson (Phase 1 Marginal Probability).

1. Suppose a water utility manager takes a random sample of 10 junctions during a pipe break. What is the probability that:

   a. The manager finds exactly 5 unsafe junctions?
   b. The manager finds 4 or more unsafe junctions?
   c. The manager finds 3 or fewer unsafe junctions?

3. Suppose now that the manager took a random sample of 20 junctions. Would it be unusual for the Manager to find 10 or more unsafe junctions? Justify your answer with a probability calculation.

3. Would you be surprised if the manager found between 8 to 10 unsafe junctions? Show your work.

4. **The "I" in BINS:** In a real water network, junctions are connected by pipes. If one junction is unsafe because of a nearby pipe break, is it truly "independent" of the junction next to it? How might a lack of independence affect the validity of using a Binomial model in a real-world engineering report?

5. **Contextual Risk:** Why is it more useful for a city official to know the distribution of unsafe junctions (Binomial) rather than just the probability that a single junction is unsafe (Marginal)?

Teacher Notes

This simulation allows you to enter the sample size (keep in mind that there are only 92 junctions in the system) as well as how many simulated trials you want to run. On each run, a random critical pipe is closed, the sample is collected, and the chlorine level is measured at that junction. A count of the unsafe junctions is recorded in the spreadsheet as shown below for each of the XX trials

	A	B	C	D
1	Trial_ID	Failed_Pipe	Sample_Size	Num_Unsafe
2	1	183	20	7
3	2	131	20	3
4	3	187	20	2
5	4	20	20	1
6	5	333	20	3
7	6	321	20	6
8	7	330	20	2
9	8	187	20	2
10	9	129	20	3
11	10	330	20	3
12	11	329	20	11
13	12	333	20	3
14	13	193	20	1
15	14	179	20	5
16	15	333	20	1
17	16	20	20	0
18	17	177	20	8
19	18	179	20	8
20	19	175	20	5
21	20	330	20	4
22	21	183	20	3
23	22	333	20	4
24	23	193	20	3
25	24	135	20	3

Summary results of the simulation are also shown:

```
==================================================
BINOMIAL SAMPLING ANALYSIS
==================================================

Sample Size (n): 20
Number of Trials: 100

Empirical Results:
  Mean number of unsafe junctions per sample: 3.9800
  Estimated probability a junction is unsafe: 0.1990
```

```
Frequency Distribution:
  0 unsafe junctions: 4 trials (0.0400)
  1 unsafe junctions: 11 trials (0.1100)
  2 unsafe junctions: 23 trials (0.2300)
  3 unsafe junctions: 22 trials (0.2200)
  4 unsafe junctions: 7 trials (0.0700)
  5 unsafe junctions: 10 trials (0.1000)
  6 unsafe junctions: 5 trials (0.0500)
  7 unsafe junctions: 3 trials (0.0300)
  8 unsafe junctions: 4 trials (0.0400)
  9 unsafe junctions: 4 trials (0.0400)
  10 unsafe junctions: 2 trials (0.0200)
  11 unsafe junctions: 2 trials (0.0200)
  12 unsafe junctions: 3 trials (0.0300)
  Estimated probability a junction is unsafe: 0.1990
```

This is intended to provide students the opportunity to build the discrete distribution (so starting with a sample of size 20 may not be ideal, maybe start with 10).

Comparing the observed distribution with a theoretical binomial distribution is part of the reason for building this simulation.

When you modify the simulation to run 1000 trials, the code modification is minimal, but also alter the %10 so there aren't too many iterations shown.

AP Statistics CED Alignment

Unit 2: Probability, Random Variables, and Probability Distributions

The following Learning Objectives (LO) and Essential Knowledge (EK) components from the 2026-27 AP Statistics Course and Exam Description are addressed in this water quality and infrastructure analysis.

Section 1: Introduction to Probability and Simulation

- **LO 2.3.A:** Estimate probabilities using simulation.
- **EK 2.3.A.1:** The probability of an event is the proportion of times the event would occur in a very long series of repetitions.
- **LO 2.4.A:** Calculate probabilities for events and their complements.
- **EK 2.4.A.3:** The Law of Large Numbers states that as the number of independent trials increases, the experimental (empirical) probability of an event approaches its theoretical probability.

Section 2: Conditional Probability and Independence

- **LO 2.1.A:** Represent the distribution of two categorical variables using a tabular representation.
- **LO 2.2.A:** Calculate summary statistics for two categorical variables.
- **EK 2.2.A.1:** Marginal and conditional relative frequencies can be calculated from a frequency table.
- **LO 2.6.A:** Calculate conditional probabilities.
- **EK 2.6.A.1:** The probability of event A, given that event B has occurred, is called a conditional probability and is denoted by P(A|B).
- **LO 2.7.A:** Determine if two events are independent.
- **EK 2.7.A.1:** Two events are independent if the occurrence of one event does not change the probability that the other event will occur.
- **EK 2.7.A.2:** Events A and B are independent if and only if P(A|B)=P(A).
- **LO 2.5.A:** Calculate probabilities for mutually exclusive events.
- **EK 2.5.A.1:** Two events are mutually exclusive (disjoint) if they cannot occur at the same time.

Section 3: Random Variables and Probability Distributions

- **LO 2.8.A:** Represent a probability distribution for a discrete random variable.
- **EK 2.8.A.1:** A probability distribution for a discrete random variable is a table, graph, or formula that gives the probability for each possible value of the random variable.
- **LO 2.9.A:** Calculate parameters for a discrete random variable.
- **EK 2.9.A.1:** The mean of a discrete random variable X, also called the expected value, is denoted by μ_X or E(X).
- **EK 2.9.A.2:** The standard deviation of a discrete random variable X is denoted by σ_X.
- **LO 2.9.B:** Interpret parameters for a discrete random variable in context.

Section 4: The Binomial Distribution

- **LO 2.10.A:** Determine if a random variable follows a binomial distribution.
- **EK 2.10.A.1:** A binomial experiment consists of n independent trials, each with two possible outcomes (success/failure) and a constant probability of success p.
- **LO 2.10.B:** Calculate probabilities for a binomial distribution.
- **EK 2.10.B.1:** The binomial probability of exactly k successes in n trials is given by $P(X=k)=\binom{n}{k}(p)^k(1-p)^{n-k}$
- **LO 2.10.C:** Calculate parameters for a binomial distribution.
- **EK 2.10.C.1:** For a binomial distribution, the mean is $\mu_X = np$ and the standard deviation is $\sigma_X = \sqrt{n \cdot p \cdot (1-p)}$

Section 5: Contextual Reasoning and Model Validity

- **LO 1.10.D:** Identify an observational study (Simulation context).
- **EK 1.10.D.5:** A confounding variable provides an alternative explanation for the observed relationship. (Relevant to the discussion of physical connectivity vs. statistical independence).
- **LO 2.7.B:** Calculate probabilities for unions of events.
- **EK 2.7.B.1:** The general addition rule: P(A∪B)=P(A)+P(B)−P(A∩B)

AP Statistics **Unit 3 Part 2– Chi Square Tests**
Sirois **Analyzing the Spread of a Contaminant from a Water Source**
Introduction to Inference for Categorical Data (Chi-Square Test for Independence)

Case Study: An Undetected (for 1 week) Contamination Event at the River Source
Background: The Physics of the Water Network
As a researcher for the Smart Water Infrastructure Modeling (SWIM) Lab, you are investigating a contamination event in the water network. The city relies on two sources. The **River (East)** is the "High-Pressure" source with a Higher Head (220.0 ft), allowing it to push water across vast distances. The **Lake (West)** is the "Low-Pressure" source with a Lower Head (167.0 ft).

The Crisis: A chemical contaminant has been introduced at the River reservoir. Because the River has the higher hydraulic energy, its contaminated water is currently dominating the flow paths, reaching junctions that the Lake's water cannot protect.

Phase 1 - Scenario: For the first 6 hours, the contaminant enters the system at the River undetected. Your goal is to determine if the physical structure of the network impacts the likelihood of a junction becoming unsafe. At hour 6, the contaminant is detected.

Introduction to the Chi-Square Model
Up to this point, we have used z-tests to compare two proportions. However, in the "River Run" event, we have **three** connection categories (1, 2, and 3+ connections). To compare multiple proportions simultaneously, we will introduce a new tool, the **Chi-Square χ^2 Test**.

The Variables:
- **Explanatory:** Connection Category (1, 2, or 3+ connections)
- **Response:** Safety Status (Safe: ≤0.05 mg/L | Unsafe: >0.05 mg/L)

Observed Counts Table

	Safe	Unsafe	Total
1 Connection	55	11	**66**
2 Connections	129	51	**180**
3+ Connections	205	101	**306**
Total	**389**	**163**	**552**

Is the safety of water in the system independent of the flow through the junction?

Before beginning our Chi Square analysis, find the following:
P(Unsafe | 1 connection)
P(Unsafe | 2 connections)
P(Unsafe | 3+ connections)

a. **Hypotheses:** Write the null and alternative hypotheses for a Chi-Square test of independence in this context.

H_0

H_A

b. **Expected Counts:** Calculate the expected counts for each cell and write them in the table below.

Formula: $E_{cell} = \dfrac{\text{row total} \times \text{column total}}{\text{overall total}}$

Expected Counts	Safe	Unsafe
1 Connection		
2 Connections		
3+ Connections		

c. **Chi-Square Components:** Calculate the component values $\chi^2_{component} = \dfrac{(o-e)^2}{e}$ for each cell.

Components	Safe	Unsafe
1 Connection		
2 Connections		
3+ Connections		

d. **Show Work:** Clearly show the calculation for the expected count of **Unsafe** junctions with **1 Connection**.

e. **Test Statistic:** $\chi^2 = \sum \dfrac{(o-e)^2}{e} =$ _____

f. **Degrees of Freedom:** df = (r−1)(c−1) = _____

g. **P-value:** Do this both by using the Chi-Square Test feature on your calculator (using matrices) and by using χ^2 cdf on your calculator. Remember, this is $P(\chi^2 > \text{test statistic} | H_0 \text{ is true})$

h. **Conclusion:** Is there evidence that water safety depends on the type of junction? Identify which specific cell(s) had the largest component values. What does this tell you about which type of junction is driving the statistical significance?

Phase 2: Closing the Pipe (Hours 6 - 12)
Event: Utility managers detect the contaminant at Hour 6 and immediately **close Pipe 60** to isolate the River Source. While the source is cut off, the "slug" of contaminated water already in the pipes continues to migrate and disperse due to existing momentum and consumer demand. The contaminated water is also in the water reservoir. Because the city shuts down the River source. No new contaminant enters. The Lake is now the only source of water. After another 6 hours of "passive recovery," the following data was collected:

Observed Counts (Hour 12):

	Safe	Unsafe	Total
1 Connection	44	22	**66**
2 Connections	104	76	**180**
3+ Connections	178	128	**306**
Total	**326**	**226**	**552**

Once again, calculate the following conditional probabilities and indicate what changes this indicates have taken place in the water network.

P(Unsafe | 1 Connection (End Node))
P(Unsafe | 2 Connections (Intermediate))
P(Unsafe | 3+ Connections (Hub))

a. **Comparison:** Compare the proportion of unsafe junctions in Phase 2 to Phase 1. Is the system getting cleaner?

b. **Conditions for Inference:** Verify the conditions for a Chi-Square test.
Random Process: ✓(Simulation randomly sampled junctions)

Independence:

Expected Counts: (Are all expected counts ≥5?)

c. **Results:** Perform the Chi-Square test (you may use your calculator).

χ^2 = _____

p-value = _____

Interpretation: Does an association between junction type and safety still exist?

Phase 3: Active Intervention (24 hours after contamination was identified)
Event: The system has been isolated for 24 hours. Without new water entering from the source, the system reaches a "stagnant state." Contamination pools in hubs and dead-ends. The response has been to clean the source of the contamination in the river and also to begin draining the contaminated water from the reservoir. No other interventions are done at this time as they prepare to get the contamination under control.

Observed Counts (24 hours after contamination identified):

	Safe	Unsafe	Total
1 Connection	39	27	**66**
2 Connections	80	100	**180**
3+ Connections	126	180	**306**
Total	**245**	**307**	**552**

a. **Risk Analysis:** In this phase, we will use **conditional probabilities** to compare risk directly.

P(Unsafe|1 Connection)=
P(Unsafe|2 Connections)=
P(Unsafe|3+ Connections)=

What seems to be happening in the system? Where at type of junction appears to have the highest proportion classified as unsafe? What explanation do you have for this phenomenon?

Phase 4: Contamination cleared, reservoir drained, pipe reopened.. (Now 48 hours after event occurred)
Event: The contamination has now been cleared so the river is clean. The reservoir has been emptied of the contaminated water. The pipe has been reopened.

Observed Counts (48 hours after contamination identified):

	Safe	Unsafe	Total
1 Connection	57	9	**66**
2 Connections	154	26	**180**
3+ Connections	257	49	**306**
Total	**468**	**84**	**552**

Analysis: Perform the Chi-Square test (you may use your calculator).

χ^2 = _____

p-value = _____

Interpretation: Does an association between junction type and safety still exist?

Do the results of your hypothesis test indicate that the water in the system is safe? Is there any type of junction that is safe?

Observational vs. Experimental Studies
In this class, we categorize studies based on how data is collected. Understanding where the Computational Modeling and Simulation (CMS) studies with EPAnet fits is important for determining what kind of conclusions we can draw. Is what we are doing an observational study or an experiment, or neither? Let's review some basics.

Observational Study: Researchers observe individuals and measure variables of interest without attempting to influence the responses.

- Generalizability: If the sample is random, we can generalize to the population.
- Causality: We cannot establish cause-and-effect due to potential confounding variables.

Experiment: Researchers deliberately impose some treatment on subjects to measure their responses.

- Generalizability: Often limited if subjects are volunteers. We often say we can generalize to the population that is similar to volunteers we recruited for the experiment.
- Causality: If subjects are randomly assigned to treatment groups, we can establish cause-and-effect.

Is CMS an Experiment?
In our water infrastructure simulation, we performed "controlled" runs. We added a contaminant, closed a pipe, emptied a reservoir, waited a fixed amount of time, opened a pipe, opened hydrants, etc. In essence, we were manipulating explanatory variables to investigate the impact they had on a response variable.

Feature	EPAnet Simulation	True Experiment
Treatment applied?	Yes (We changed pipe status/flushing)	Yes (Researcher imposes treatment)
Control?	High (We held the network constant)	High (Control groups/extraneous variables)
Random Assignment	**No**	**Yes** (Crucial for balancing groups with respect to potential confounding variables)

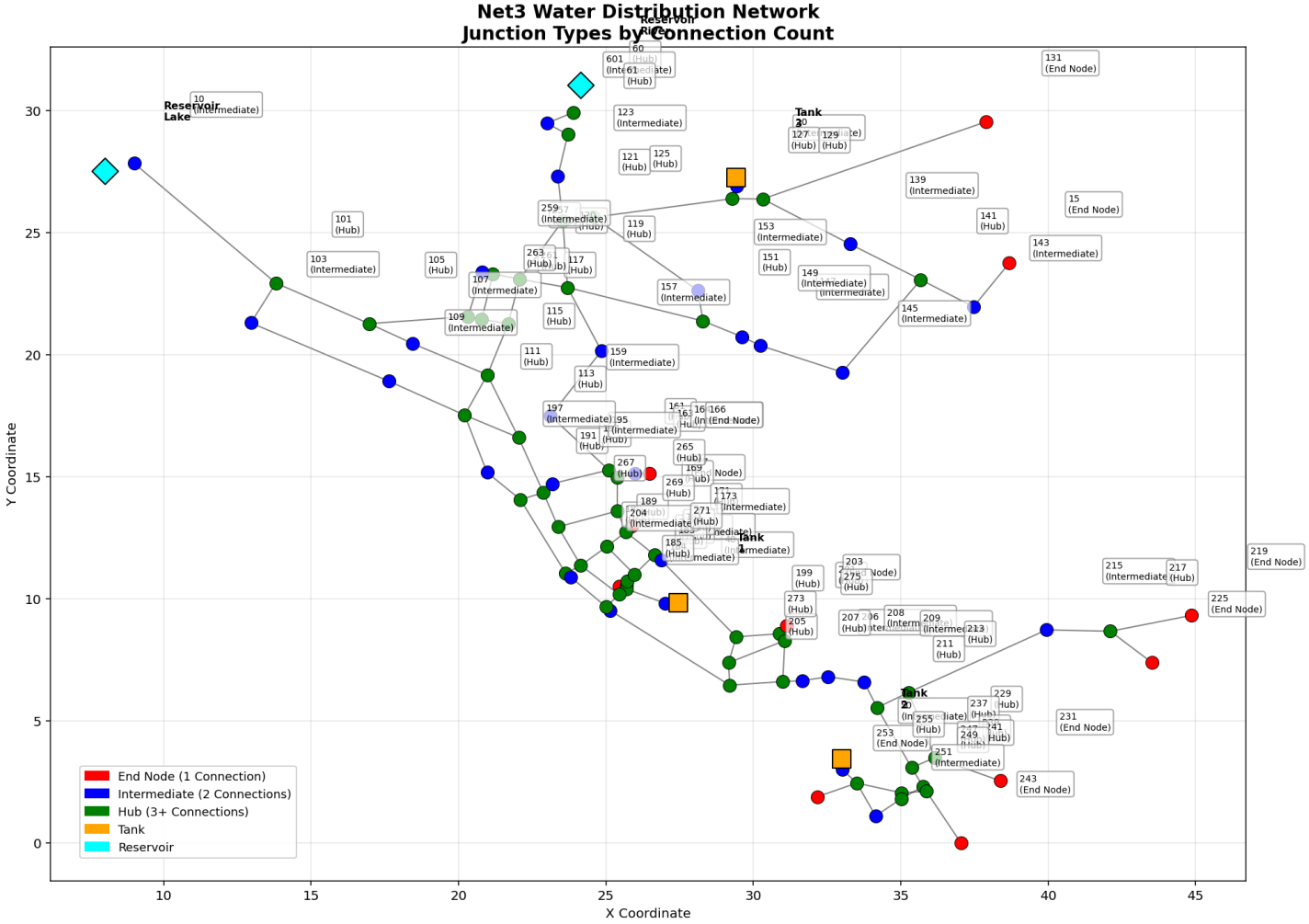
Our simulation behaves in some ways like an experiment, but it is truly a controlled modeling scenario. While it mimics the structure of an experiment, it lacks the **random assignment** of individual junctions to different treatment groups. It is also a simulation, and we need to remember that while "all models are wrong, but some are useful" (George Box famous statistician). The goal of our model is not perfection, it is to help us understand real world water infrastructure dynamics.

Establishing Cause and Effect
Can we say that "Opening the hydrants caused the water to become safe"?
Yes, within our model. Because the simulation is built on the laws of physics (hydraulics and mass balance), we can establish what is known as mechanistic cause-and-effect. We know that the change in the "Safety Status" variable was a direct result of the "Flushing" variable because every other variable in the computer code was held constant.

When extending this to the real world, we must be cautious extending these cause-effect results. A computer model is a simplification of reality. In a real city we have:

1. **Confounding Variables:** Pipe corrosion, varying consumer demand, and water temperature could all influence safety.
2. **Lack of Randomization:** We cannot randomly assign one street to be "flushed" and another to be "contaminated" to compare them fairly.

When you consider our analysis from today's activity and the Chi-Square results from each phase, remember that CMS allows us to test "What if" scenarios that would be too dangerous, expensive or unethical in the real-world.


Net3 Water Distribution Network
Junction Types by Connection Count
X Coordinate
Y Coordinate
End Node (1 Connection)
Intermediate (2 Connections)
Hub (3+ Connections)
Tank
Reservoir
Reservoir Lake
Tank 1
Tank 2

Teacher Notes

Here is a high-level all in one place overview of the rather complicated scenario.

The simulation models a municipal water distribution network fed by two sources: the River (High-Pressure/East) and the Lake (Low-Pressure/West). This scenario investigates a chemical contamination event at the River reservoir (introduced at 10 mg/L) and the subsequent engineering interventions.

The Five-Phase Timeline

- **Phase 1 (Hours 1–6):** Contaminant enters the River source undetected and spreads through the network.
- **Phase 2 (Hours 7–12):** Detection occurs; the primary transmission pipe (Pipe 60) is closed to isolate the source.
- **Phase 3 (Hours 19–24):** Long-term closure leads to a "stagnant state" where contamination pools in specific areas.
- **Phase 4 (Hours 25–48):** The River source is cleaned, and the reservoir is drained. The pipe is reopened, allowing for natural mixing and recovery.
- **Phase 5 (Hours 49–384):** An extended 2-week recovery period involving staged hydrant flushing to clear residual contaminants from dead-ends and hubs.

Data Classification & Variables

Explanatory Variable: Junction Type

Junctions are classified by their connection count (structural position):

1 Connection (End Node): The "dead-ends" or cul-de-sacs of the system.

2 Connections (Intermediate): Standard pass-through intersections.

3+ Connections (Hub): Major intersections where multiple pipes meet.

Response Variable: Safety Status

Safety is determined by the contaminant concentration (mg/L) at specific sampled timestamps:

- **Safe:** Concentration ≤0.5 mg/L
- **Unsafe:** Concentration >0.5 mg/L

Raw data is provided to once again provide students with the opportunity to work (a little) with the data to classify into a contingency table. Don't spend too much time on this, but do have them see how this type of data is collected and how the contingency table organizes it.

Each trial will produce all the contingency tables you need if you want to run it in front of students and just work with what is provided at each phase.

This provides an opportunity to discuss why end nodes may have a lower probability of being unsafe, and other concepts. Data is collected in 6x1 hour blocks for the entire simulation. This was done to make sure conditions for Chi Square tests were met in terms of expected counts.

	A	B	C	D	E	F
1	Phase	Junction_ID	Junction_Type	Hour	Concentration_mgL	Safety_Status
2	Phase 1	10	2 Connections (Intermediate)	1	0	Safe
3	Phase 1	10	2 Connections (Intermediate)	2	0	Safe
4	Phase 1	10	2 Connections (Intermediate)	3	0	Safe
5	Phase 1	10	2 Connections (Intermediate)	4	0	Safe
6	Phase 1	10	2 Connections (Intermediate)	5	0	Safe
7	Phase 1	10	2 Connections (Intermediate)	6	0	Safe
8	Phase 1	15	1 Connection (End Node)	1	0	Safe
9	Phase 1	15	1 Connection (End Node)	2	0	Safe
10	Phase 1	15	1 Connection (End Node)	3	0	Safe
11	Phase 1	15	1 Connection (End Node)	4	0	Safe
12	Phase 1	15	1 Connection (End Node)	5	6.57654	Unsafe
13	Phase 1	15	1 Connection (End Node)	6	8.959362	Unsafe
14	Phase 1	20	2 Connections (Intermediate)	1	0	Safe
15	Phase 1	20	2 Connections (Intermediate)	2	0	Safe
16	Phase 1	20	2 Connections (Intermediate)	3	10	Unsafe
17	Phase 1	20	2 Connections (Intermediate)	4	10	Unsafe
18	Phase 1	20	2 Connections (Intermediate)	5	10	Unsafe
19	Phase 1	20	2 Connections (Intermediate)	6	10	Unsafe
20	Phase 1	35	1 Connection (End Node)	1	0	Safe
21	Phase 1	35	1 Connection (End Node)	2	0	Safe
22	Phase 1	35	1 Connection (End Node)	3	0.004989	Safe
23	Phase 1	35	1 Connection (End Node)	4	9.591794	Unsafe
24	Phase 1	35	1 Connection (End Node)	5	9.183358	Unsafe
25	Phase 1	35	1 Connection (End Node)	6	8.374804	Unsafe

AP Statistics CED Alignment
Unit 3 Part 2: Contaminant Spread Chi Square Analysis Lesson

This document summarizes the alignment between the "Analyzing the Spread of a Contaminant from a Water Source" case study and the AP Statistics Learning Objectives (LO) and Essential Knowledge (EK) standards.

Unit 3: Inference for Categorical Data: Proportions
LO 3.14: Setting Up a Chi-Square Test for Homogeneity or Independence
EK 3.14.1: Students formulate null and alternative hypotheses for a Chi-Square test of independence.
EK 3.14.2: Students identify the appropriate test based on the data collection method (one sample, two variables).

LO 3.15: Carrying Out a Chi-Square Test for Homogeneity or Independence
EK 3.15.1: Students calculate expected counts, chi-square components, and the test statistic.
EK 3.15.2: Students determine degrees of freedom and p-values.
EK 3.15.3: Students interpret the p-value and make a conclusion in context.

Unit 2: Probability, Random Variables, and Probability Distributions
LO 2.6: Calculate Conditional Probabilities
EK 2.6.1: Students calculate conditional probabilities from a two-way table.

Unit 1: Exploring One-Variable Data and Collecting Data
LO 1.2.B: Identify Types of Variables
EK 1.2.B.1: Students identify categorical variables.

LO 1.3.A: Construct Categorical One-Variable Tabular Representations
EK 1.3.A.1: Students work with frequency tables.

LO 1.10.D: Identify an Observational Study
EK 1.10.D.1: Students distinguish between observational studies and experiments.

Statistical Practices Addressed
Practice 1: Formulate Questions
- **Skill 1.A:** Determine a valid investigative question. Students evaluate if water safety depends on the physical flow through a junction.

Practice 2: Collect Data
- **Skill 2.C:** Identify appropriate statistical inference methods. Students must recognize that a Chi-Square test is required for a 3×2 table.

Practice 3: Analyze Data
- **Skill 3.C:** Calculate expected counts and probabilities.
- **Skill 3.E:** Calculate appropriate statistical inference method results (Test Statistic, P-value).

Practice 4: Interpret Results
- **Skill 4.B:** Justify a claim based on statistical calculations.
- **Skill 4.F:** Interpret results of statistical inference methods.
- **Skill 4.G:** Justify a claim based on inference results (Conclusion in context).

Summary of Overall Alignment
The worksheet provides a comprehensive application of **Inference for Categorical Data**. By using a longitudinal simulation (Phases 1–4), it forces students to move beyond rote calculation and into **Statistical Practice 4 (Interpret Results)**. Students must explain why certain junction types (Hubs vs. End Nodes) drive the statistical significance, the goal is to both make a connection and to bridge a gap between mathematical output and physical infrastructure modeling.

AP Statistics **Regression Analysis**
Sirois **Analyzing Relationships Between Quantitative Variables**
Exploring Relationships Under Stress: Normal vs. Failure Conditions

Revisiting The Physics of the Water Network
In our previous SWIM Lab investigations, we explored how a "Critical Pipe Failure" forces water to take circuitous detours, increasing Water Age and causing Chlorine Decay. We also know that Elevation plays a primary role in determining Hydraulic Pressure.

Today, we will use Linear Regression to model these relationships. We will compare the system under Day 1 (Normal Operation) and Day 2 (Stress Event: After a Critical Pipe is Closed).

Part A: Water Quality Analysis (Water Age vs. Chlorine)

The Investigative Question: Is there a linear relationship between Water Age (hours) and Chlorine Concentration (mg/L) in the water network, and does this relationship change during a stress event?

Before beginning with regression analysis, ask yourself if the investigative question contains the components necessary to guide a statistical investigation. Discuss within your group.

Graphical Exploration: Open the Lesson 8 Regression Analysis Data.xlsx file. Open the Day 1 data sheet. Construct a scatterplot of Chlorine (y) vs. Water Age (x) in Desmos

a. Describe the relationship (Strength, Direction, Form, Unusual Features):

b. Does a linear model seem appropriate? Justify your answer.

LSRL Equation: Use technology to calculate the equation of the Least Squares Regression Line (LSRL) as well as the other statistical measures (correlation, residuals) that come along with the LSRL.

- LSRL Equation: _____
- Interpret the **slope** in context:
- Interpret the **y-intercept** in context. Does it have a meaningful physical interpretation?

Correlation and Determination:
- Interpret r (correlation coefficient) in context:
- Interpret r2 (coefficient of determination) in context:

Residual Analysis: In Desmos, when you calculate the LSRL, residuals are automatically calculated.

Remember that a residual = actual value – predicted value

Calculate the residual for one of the junctions in the data set. Show your work below and explain what that residual indicates about the prediction in comparison with the actual value. As always, write your answer in context.

If the residual is **positive**, what does that mean about the actual chlorine level compared to the model's prediction?

Predictions:

- **Interpolation:** Predict the chlorine level for a junction with a Water Age of 15 hours.
- **Extrapolation:** Predict the chlorine level for a junction with a Water Age of 60 hours. (Note: Day 1 data only goes up to 40 hours). How confident are you in this prediction?

Part B: Hydraulic Analysis (Elevation vs. Pressure)
The Investigative Question: How does Elevation (ft) influence Hydraulic Pressure (psi) across the 92 junctions of the water system?

The Day 1 Model (Normal): Construct a scatterplot and find the LSRL for Pressure (y) vs. Elevation (x).

LSRL Equation: _____

Interpret the **slope**, in context:

Describe the relationship between elevation and pressure in the water network. Be sure to include strength, direction, form and any unusual features that may be evident.

The Day 2 Model (Stress Event):
Now open the tab for Day 2 Day2_Stress_Regression_Data.xlsx. This data was collected after a critical pipe was closed.

LSRL Equation: _____

Compare the Day 2 slope and R^2 to Day 1. What happened to the relationship?

The Danger of Extrapolation and Combined Data

The Combined View: Construct a scatterplot using the **Combined Data** sheet (Day 1 and Day 2 data together). Remember that the Day 1 sheet contains hours 0 through 24, the Day 2 sheet contains hours 24 through 48, but fresh water may be entering the system. From viewing the scatterplot, what is the domain of the data you are studying. Also, the Combined sheet contains hours 0 through 48. Also consider there are no elevations lower than 0 or higher than 45.

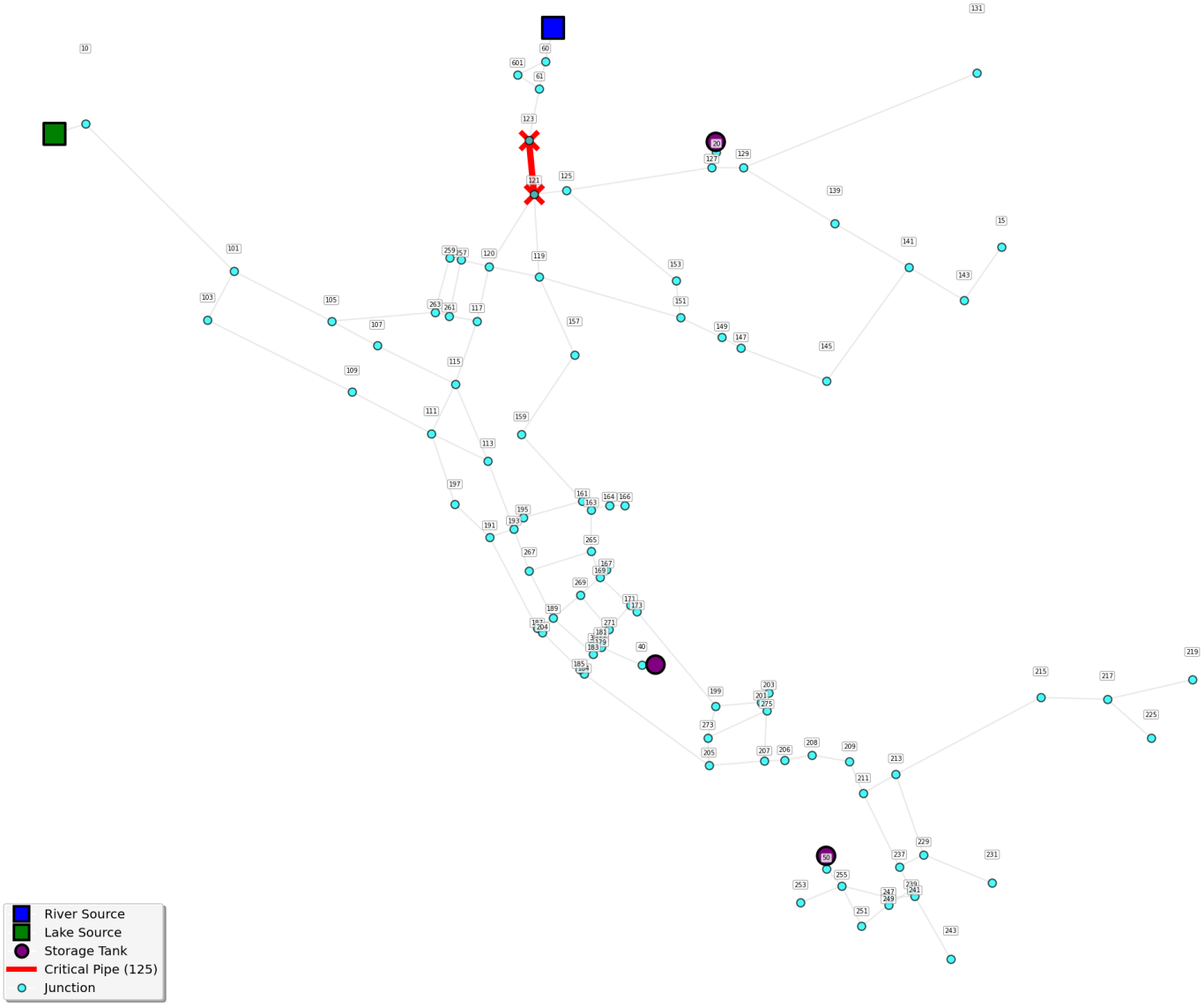
Describe what you see. When you look at the combined scatterplot, do you see evidence of two distinct linear patterns? If so, describe how they differ.

Extrapolation Challenge: Use your Day 1 (Normal) Pressure model to predict the pressure for a junction at 50 ft elevation during the Day 2 (Stress) event.

Compare this prediction to the actual Day 2 data for that elevation.

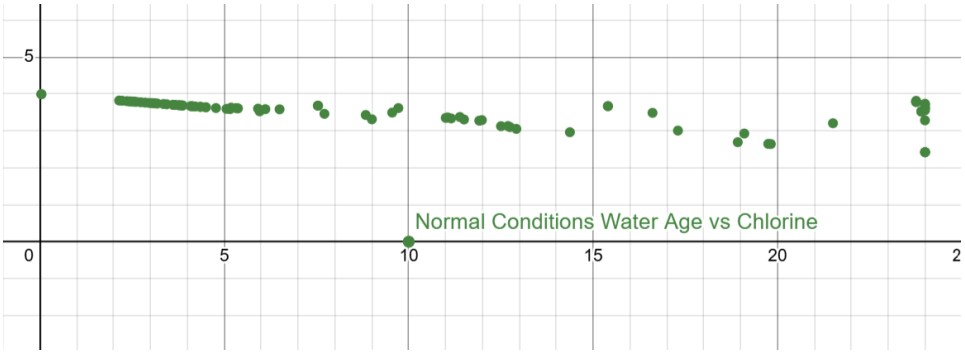
Conclusion: Why is it dangerous to use a model developed under one set of conditions (Normal) to make predictions for a different set of conditions (Stress)?

Net3 Water Network: Linear Regression Analysis
Critical Pipe: 125



Teacher Notes Graphs to Expect

Day 1 Normal Age vs Chlorine



$c_1 \sim m \cdot a_1 + b$

REGRESSION PARAMETERS

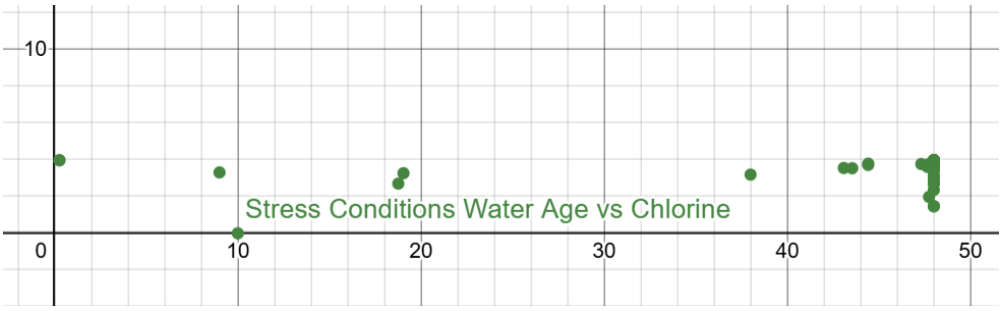
$m = -0.0228959$ $b = 3.755$

STATISTICS RESIDUALS

$R^2 = 0.2883$ e_1

$r = -0.5369$

Day 2 Stress Age vs Chlorine



$c_1 \sim m \cdot a_1 + b$

REGRESSION PARAMETERS

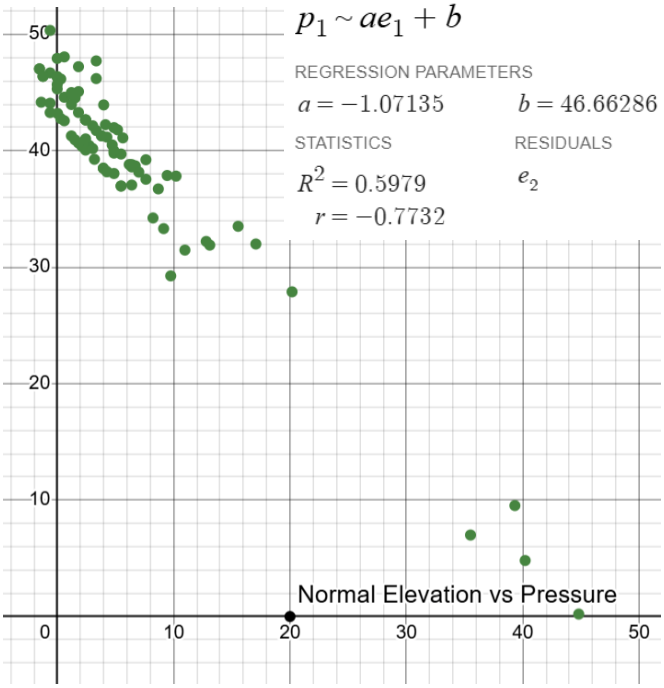
$m = 0.00454294$ $b = 3.48115$

STATISTICS RESIDUALS

$R^2 = 0.006915$ e_1

$r = 0.08315$

Elevation vs Pressure (Normal Day 1)



$p_1 \sim ae_1 + b$

REGRESSION PARAMETERS

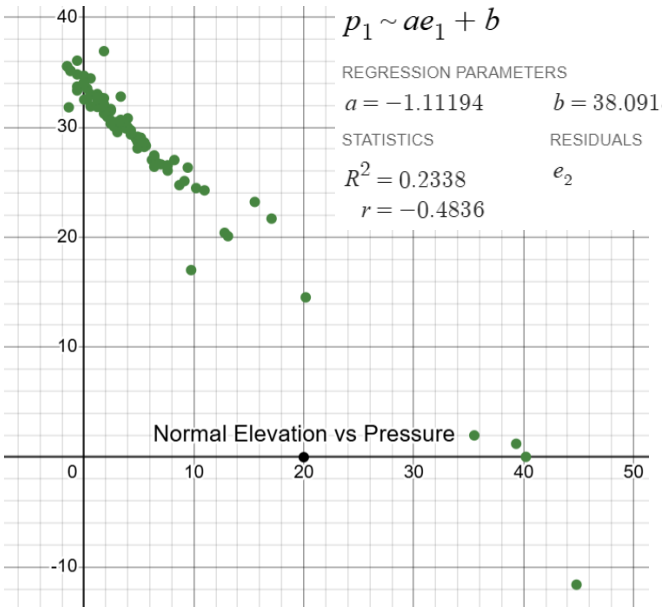
$a = -1.07135$ $b = 46.66286$

STATISTICS RESIDUALS

$R^2 = 0.5979$ e_2

$r = -0.7732$

Elevation vs Pressure (Stress Day 2)



$p_1 \sim ae_1 + b$

REGRESSION PARAMETERS

$a = -1.11194$ $b = 38.09189$

STATISTICS RESIDUALS

$R^2 = 0.2338$ e_2

$r = -0.4836$

Notice negative pressure reading on Junction 10. In this system Junction 10 is the highest elevation. Negative pressure means water is being drawn faster than it can be supplied, so this is a system failure.

AP Statistics CED Alignment: Unit 5 Regression Analysis

The following Learning Objectives (LO) and Essential Knowledge (EK) components from the 2026-27 AP Statistics Course and Exam Description are addressed in this lesson.

Section 1: Linear Regression Models

- **LO 5.1.A:** Construct graphical representations between two quantitative variables.
  - EK 5.1.A.1: A scatterplot is used to represent the relationship between two quantitative variables.
- **LO 5.2.A:** Calculate and interpret the correlation coefficient (r).
  - EK 5.2.A.1: The correlation coefficient (r) measures the direction and strength of the linear relationship.
- **LO 5.3.A:** Calculate and interpret a predicted response using a linear regression model.
  - EK 5.3.A.2: Extrapolation is the use of a regression line for prediction far outside the interval of values of the explanatory variable used to obtain the line. Such predictions are often not accurate.

Section 2: Residuals and Least-Squares

- **LO 5.4.A:** Calculate and interpret residuals.
  - EK 5.4.A.1: A residual is the difference between an observed value of the response variable and the value as predicted by the regression line (observed – predicted).
- **LO 5.5.A:** Interpret the slope and y-intercept of a least-squares regression line.
  - EK 5.5.A.1: The slope (b) of a regression line describes the predicted change in the response variable for each unit increase in the explanatory variable.
  - EK 5.5.A.2: The y-intercept (a) is the predicted value of the response variable when the explanatory variable is equal to 0.

Section 3: Coefficient of Determination

- **LO 5.5.B:** Calculate and interpret the coefficient of determination (R^2).
  - EK 5.5.B.1: The coefficient of determination (R^2) measures the percentage of total variation in the response variable that is explained by the least-squares regression line.

Statistical Practices Addressed

- **Practice 3 (Analyze Data):** Students calculate LSRL, r, R^2, and residuals using SWIM Lab datasets.
- **Practice 4 (Interpret Results):** Students justify why a model built on "Normal" data fails to predict "Stress" conditions, illustrating the physical and statistical limits of extrapolation.